Towards Scalable and Generalizable Video Retrieval

by

Zhuoning Guo

A Thesis Submitted to
The Hong Kong University of Science and Technology (Guangzhou)
in Partial Fulfillment of the Requirements for
the Degree of Doctor of Philosophy
in Artificial Intelligence

July 2026, Guangzhou

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This is to certify that I have examined the above PhD thesis
and have found that it is complete and satisfactory in all respects,
and that any and all revisions required by
the thesis examination committee have been made.

Prof. Xiaowen Chu and Prof. Renjing Xu, Thesis Supervisor

Prof. Xiaowen Chu, Head of Thrust

Thrust of Artificial Intelligence

29 July 2026

ACKNOWLEDGMENTS

I would like to express my sincere gratitude to my supervisors, Prof. Xiaowen Chu and Prof. Renjing Xu, for their invaluable guidance, continuous support, and encouragement throughout my PhD journey. Their expertise, vision, and patience have been instrumental in shaping this work.

I am deeply grateful to all committee members for their insightful feedback and constructive suggestions that have greatly improved this thesis.

I wish to thank my colleagues and collaborators in the Thrust of Artificial Intelligence at HKUST(GZ) for the stimulating discussions, collaborative spirit, and friendship that made my research journey rewarding.

My heartfelt appreciation goes to my family and friends for their unwavering support, understanding, and love throughout these years.

Finally, I acknowledge the generous support from the Hong Kong University of Science and Technology (Guangzhou) for providing excellent research facilities and resources.

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Zhuoning Guo

Thrust of Artificial Intelligence

The Hong Kong University of Science and Technology (Guangzhou)

ABSTRACT

Video retrieval is a foundational operation in modern multimedia search, recommendation, and content discovery systems. Given a user query, a retrieval system must identify relevant videos from a large candidate collection and provide reliable candidates for downstream ranking, recommendation, or user interaction. As online video collections continue to grow in scale, duration, modality, and semantic complexity, video retrieval can no longer be adequately characterized as short-clip matching under a fixed benchmark protocol. Practical retrieval systems must support heterogeneous query forms, recognize evidence at different semantic granularities, construct dependable video representations from limited visual observations, and search large candidate spaces under stringent latency and throughput constraints.

This thesis studies embedding-based video retrieval as a scaling problem along the retrieval pipeline. Building a retrieval system involves three decisions in sequence, each pre-

senting a distinct scaling challenge: what semantic capabilities the model should possess and how to scale toward them, how visual observation should adapt to videos of varying length and query granularity, and how learned retrieval signals can scale to large candidate collections where metric design, approximate search, and hardware execution are tightly coupled. We term these three challenges semantic scaling, observational scaling, and search scaling. Concretely, this thesis develops three complementary artifacts: a general video embedder trained through capability-guided synthesis, a multi-granular long-video retrieval benchmark together with an efficient inference framework, and a GPU-accelerated graph retrieval system deployed at web scale. Across the three stages studied in this thesis, the results show that scaling model capacity, observation density, or metric accuracy alone is insufficient for broader retrieval generalization unless it is aligned with the structure of the corresponding bottleneck. Scalable video retrieval therefore requires structure-aware scaling across semantic capability, visual observation, and search execution. Within this thesis, generalizability serves as an empirical criterion for assessing whether such scaling has been performed effectively at each stage. By addressing these challenges in sequence, the thesis connects representation learning, observation-aware embedding generation, and deployment-scale retrieval for building video retrieval systems that are both scalable and generalizable.

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Video retrieval is a central task in multimedia information systems, supporting content search, recommendation, personalization, and large-scale video discovery. Given a query, a retrieval system identifies relevant videos from a candidate collection and provides a compact set of candidates for subsequent ranking or decision-making [1]. Early research formulated this as text-to-video matching over short clips, evaluated by the alignment between text queries and corresponding clips. The field has progressed from adapting Contrastive Language-Image Pretraining (CLIP) [2] to video through temporal modules [3] to employing Multimodal Large Language Models (MLLMs) that offer superior language understanding and visual generalization [4]. These models are typically trained on massive datasets with noisy text annotations [5] and evaluated on standard benchmarks such as MSR-VTT [6], achieving strong results for coarse-grained text-to-video retrieval. In parallel, retrieval-and-ranking architectures [7, 8] have become the standard deployment paradigm for video search and recommendation systems. Contemporary video retrieval, however, extends well beyond this controlled regime. Retrieval systems increasingly encounter heterogeneous query forms, richer semantic conditions, longer and more informationally dense videos, and candidate collections whose size precludes exhaustive scoring. These developments shift video retrieval from a fixed benchmark matching problem to a broader methodological question: how can retrieval scale beyond the controlled conditions under which it was originally developed?

Building an embedding-based video retrieval system involves three decisions in sequence, each presenting a distinct scaling challenge. First, the semantic capacity of the retrieval model must scale to the diversity of queries and relevance patterns encountered in practice. Second, the observation strategy through which visual evidence is provided to the model must scale to videos of varying length and information density. Third, the learned retrieval signal must scale to large candidate collections while satisfying the latency and throughput constraints of real serving systems. We term these three challenges semantic scaling, observational scaling, and search scaling. They follow the information flow of the retrieval pipeline, addressing successive stages from representation to deployment. Yet each presents a distinct scaling bottleneck that is not eliminated by optimizing an upstream stage alone. A better representation does not automatically determine how videos should be observed, and a better observation strategy does not automatically make retrieval signals searchable over billion-scale collections. Addressing scalability in video retrieval therefore requires understanding the specific bottleneck at each stage and designing solutions targeted at its structure.

A first challenge is that retrieval representations must cover increasingly diverse relevance conditions, including heterogeneous query modalities, fine-grained visual details, temporal dynamics, and partial relevance within long videos. We refer to this problem as semantic scaling. However, the evaluation landscape is fragmented. Recent benchmarks have advanced specific aspects such as spatiotemporal grounding [9], composed retrieval [10], and long-context video understanding [11], but models are optimized and tested on isolated benchmarks, preventing a clear understanding of their comprehensive capabilities. This fragmentation obscures a deeper problem. Performance on conventional benchmarks correlates poorly with holistic retrieval capability, and scaling model size alone does not resolve these deficiencies. Image-only trained models [12] can perform surprisingly well on video retrieval tasks, and compact models can outperform much larger ones on specific retrieval

scenarios [13], suggesting that expanding model parameters and data volume does not by itself yield broader retrieval capability. Two additional gaps compound the problem: there is no systematic diagnostic framework for identifying which capabilities a model is missing, and existing data synthesis approaches [14, 15] produce resources with uneven quality and distribution. Without knowing which capabilities to scale toward, representation learning expands in directions that may not address the most consequential deficiencies.

A second challenge is that video embeddings are generated from limited visual observations rather than complete video content. We refer to the problem of scaling visual evidence acquisition across video length, information density, and query granularity as observational scaling. Before a video can be processed by an embedding model, it must be converted into a discrete sequence of visual inputs through spatiotemporal sampling. Since encoding every frame at full resolution is computationally prohibitive [16], this sampling step fundamentally determines the visual evidence available to the representation [17]. Too few frames or too low a resolution can cause the target event to be missed entirely, constituting a structural information bottleneck that cannot be addressed by representation learning alone. Yet most existing retrieval research operates in settings where this constraint is invisible. Short-clip datasets such as MSR-VTT [6] and DiDeMo [18] are brief enough that sparse sampling captures most of the video content. Real-world videos, however, are increasingly long and informationally dense [19, 20], and user queries may describe anything from the overall theme to a brief, specific moment, making retrieval inherently multi-granular [21, 22]. How sampling density interacts with query granularity is a central question for long-video retrieval [23], yet this relationship remains underexplored because no existing benchmark provides the required testbed. Short-clip benchmarks are too brief for sampling density to matter, long-video benchmarks [11] evaluate at a single query granularity, and benchmarks with fine-grained temporal annotations feature videos that are relatively short. Without a setting that combines long-form visual evidence with queries at multiple granularity lev-

els, the scaling behavior of retrieval with respect to observation remains poorly understood. Recent work has begun to study scaling laws for frame count and caption density on short clips [24], but the interaction between spatiotemporal sampling and retrieval granularity on long-form video has not been investigated.

A third challenge is that learned retrieval signals must remain searchable under large-corpus and serving constraints. We refer to this deployment-stage bottleneck as search scaling. Video retrieval and recommendation systems in production operate over corpora that may contain hundreds of millions to billions of items [25, 26], where exhaustive scoring is incompatible with online latency requirements [27]. Graph-based approximate nearest neighbor search methods such as HNSW [28] have become the dominant approach, but a structural conflict exists between learned retrieval metrics and graph-based search. Neural relevance metrics often violate the triangle inequality [29], the property that ensures optimal pathfinding during graph traversal, forcing the exploration of redundant edges and causing greedy traversal to become trapped in local regions of the graph. Beyond metric-search incompatibility, high concurrency introduces hardware execution bottlenecks. Divergent queries trigger different search paths, producing irregular neural computations that lead to frequent GPU kernel launches with fixed latency overhead dominating runtime [30, 31]. These challenges are tightly coupled: the learned metric affects graph traversal behavior, the traversal algorithm determines computational patterns, and the hardware execution strategy constrains achievable throughput. Optimizing any single component in isolation does not resolve the overall system bottleneck.

To address these three challenges, the thesis develops targeted solutions at successive stages of the retrieval pipeline. The first study addresses semantic scaling through capability diagnosis, targeted data synthesis, and curriculum-based training of a general video embedder. The second study addresses observational scaling by introducing a multi-granular long-video retrieval setting, analyzing spatiotemporal scaling behavior across model families,

and developing a dense-to-sparse inference framework. The third study addresses search scaling by co-designing learned relevance metrics, graph traversal, and GPU execution for deployment-scale retrieval.

Together, these studies address successive bottlenecks along the retrieval pipeline, from semantic representation to visual observation and large-scale search execution. Across the three stages studied in this thesis, the results show that scaling model capacity, observation density, or metric accuracy alone is insufficient for broader retrieval generalization unless it is aligned with the structure of the corresponding bottleneck. Scalable video retrieval therefore requires structure-aware scaling at each stage. Within this thesis, generalizability serves as an empirical criterion for assessing whether such scaling has been performed effectively. By addressing semantic, observational, and search scaling in sequence, this thesis develops a methodology for video retrieval that extends from representation learning through observation-aware embedding generation to deployment-scale search, providing a path toward retrieval systems that are both scalable and generalizable.

1.2 Organization of the Thesis

The remainder of this thesis is organized as follows.

Chapter 2 reviews the background and related literature, covering fundamental concepts and prior work on video embedding models, benchmarks and evaluation protocols, and large-scale retrieval. Chapter 3 addresses semantic scaling through capability-guided evaluation, targeted data synthesis, and curriculum learning, producing the General Video Embedder (GVE). This study establishes the representation foundation that the subsequent studies build upon. Chapter 4 addresses observational scaling by introducing the Multi-Granular Long Video Retrieval (MG-LVR) benchmark, analyzing spatiotemporal scaling behavior, and developing the Video Embedding Waterfall (VEterfall) framework for effi-

cient inference. This study extends the investigation from what the model should represent to how visual evidence is provided to the model. Chapter 5 addresses search scaling through co-designed learned metrics, parallel graph search (HiPANNS), and GPU-accelerated execution, validated through deployment in a large-scale video recommendation system. This study addresses the final stage of the retrieval pipeline. Chapter 6 summarizes the main contributions and discusses future directions. Chapter 7 lists the publications associated with this thesis.

CHAPTER 2

BACKGROUND AND LITERATURE REVIEW

This chapter provides the background and surveys the literature relevant to this thesis. §2.1 introduces the formal concepts and problem formulations underlying video retrieval, organized along the retrieval pipeline: problem definition, embedding models, spatiotemporal preprocessing, and large-scale search. §2.2 surveys existing work along three research themes that correspond to the three contribution chapters: video embedding models and their training paradigms (Chapter 3), benchmarks and evaluation protocols (Chapters 3 and 4), and large-scale retrieval systems (Chapter 5).

2.1 Basic Concepts

This section introduces the formal concepts and problem formulations underlying this thesis, organized along the core pipeline of video retrieval: problem formulation (§2.1.1), embedding models (§2.1.2), spatiotemporal preprocessing (§2.1.3), and large-scale search (§2.1.4).

2.1.1 Video Retrieval: Problem Formulation

Video retrieval is the task of identifying videos from a collection that satisfy a user's information need. Given a query q and a video corpus $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ of N candidate videos, the goal is to retrieve the subset of videos whose content is relevant to q [1]. In the ranking formulation, the system produces an ordered list of videos sorted by their estimated relevance scores $s(q, v_i)$.

Video retrieval systems support diverse query formats:

- **Textual Video Retrieval** (a.k.a., Text-to-Video Retrieval). The query q is a natural language description, and the system retrieves videos matching the textual description. This is the most widely studied formulation.

- **Composed Video Retrieval**. The query q combines multiple modalities, typically a text description and a reference image or video, supporting scenarios where users provide both visual examples and textual specifications to refine their search intent.

- **Visual Video Retrieval**. The query q is a pure visual input (image or video), and the system retrieves semantically similar videos from the corpus.

Beyond query modality, video retrieval tasks vary in the granularity and nature of the semantic match required:

- **Coarse-grained retrieval** matches videos at the level of overall themes, topics, or general content categories (e.g., "a cooking tutorial"). The primary challenge lies in semantic abstraction: the system must capture the gist of a video rather than attend to specific visual details.

- **Fine-grained retrieval** requires precise understanding of detailed visual attributes, spatial relationships, temporal dynamics, or camera movements within short clips (e.g., "a person slicing onions with a ceramic knife on a wooden board"). The challenge is perceptual precision: distinguishing visually similar but semantically distinct content.

- **Partially relevant retrieval** addresses scenarios where only a short segment of a video is relevant to the query, while the remainder is irrelevant. The core challenge is temporal localization under a whole-video representation: the system must identify the matching fragment without being misled by the dominant but irrelevant content.

- **Long-context retrieval** involves queries that require holistic understanding of extended temporal context (such as narrative progression, causal chains, or cross-segment reasoning) rather than matching a single localized event. The challenge is temporal integration: the relevant evidence is distributed across the video and cannot be reduced to any individual segment.

- **Long-video retrieval** extends the temporal scope to hour-long videos (e.g., movies, lectures, livestreams), where relevant content may occupy only a tiny fraction of the total duration. The challenge is temporal sparsity at scale: efficiently identifying sparse relevant signals within extremely long sequences without exhaustive processing.

Although these five domains may appear similar in form, all mapping a query to a ranked list of videos, they pose fundamentally different challenges: coarse-grained retrieval tests semantic abstraction, fine-grained retrieval tests perceptual precision, partially relevant retrieval tests temporal localization, long-context retrieval tests temporal integration, and long-video retrieval tests scalability under temporal sparsity. This taxonomy of query modalities and semantic domains collectively defines the **semantic scale** of video retrieval: the breadth of relevance conditions that a retrieval representation must support. A general-purpose retrieval model should perform effectively across this diverse space rather than optimize for a single matching pattern.

2.1.2 Video Embedding Models

Modern video retrieval systems rely on embedding-based approaches, where queries and videos are mapped to dense vector representations in a shared semantic space, reducing retrieval to nearest neighbor search in the embedding space.

Let $E_q : \mathcal{Q} \to \mathbb{R}^d$ and $E_v : \mathcal{V} \to \mathbb{R}^d$ denote the query and video embedding functions,

which map inputs to d-dimensional vectors. The relevance between a query q and a video v is computed as the cosine similarity between their embeddings:

$$s(q, v) = \frac{E_q(q) \cdot E_v(v)}{\|E_q(q)\| \, \|E_v(v)\|} \tag{2.1}$$

The embedding functions are implemented as neural network encoders. Depending on the underlying architecture, these encoders can be broadly categorized into two families: those built upon CLIP [2], which employ a dual-encoder architecture with modality-specific encoders for text and video respectively, and those based on Multimodal Large Language Models (MLLMs), which adopt a unified architecture that processes text, images, and video through the same model. The architectural distinctions and representative models of each family are discussed in detail in §2.2.1. The challenge of training such models for heterogeneous retrieval conditions is addressed in Chapter 3.

2.1.3 Spatiotemporal Sampling

Before a video can be processed by an embedding model, it must be converted into a discrete sequence of visual inputs through spatiotemporal sampling. Since encoding every frame at full resolution is computationally prohibitive [16], this preprocessing step critically determines both the available visual evidence and the retrieval quality [17].

Temporal sampling selects F frames from a video of duration T. The most common strategy is uniform sampling, where frames are evenly spaced across the video duration.

Spatial sampling determines the visual resolution or token representation per frame. For ViT-based architectures, this involves converting frames into sequences of visual tokens, where the token count S per frame determines the spatial detail preserved.

The combination of temporal frame count F and spatial token count S per frame defines the total input size $F \times S$ for the embedding model. This decomposition into orthogonal

spatiotemporal dimensions allows systems to trade off temporal coverage against spatial detail under a fixed computational budget [24].

The sampling configuration used during model training, denoted $(F_{\text{train}}, S_{\text{train}})$, may differ from that used during inference, denoted $(F_{\text{test}}, S_{\text{test}})$. In practice, inference often operates under stricter computational constraints, leading to a density mismatch between training and deployment. The impact of this mismatch on retrieval quality, and the extent to which retrieval performance depends on query granularity, are central questions in the **observational scale** of video retrieval, the fidelity and selectivity with which visual evidence is preserved for embedding generation.

2.1.4 Large-Scale Search: ANNS and Retrieval Systems

When a video retrieval system is deployed at scale, the candidate corpus may contain millions to billions of videos. Exhaustive similarity computation between a query embedding and every candidate becomes computationally prohibitive, motivating the development of **Approximate Nearest Neighbor Search (ANNS)** algorithms that achieve sub-linear query time by building index structures for selective exploration of the database.

Three main categories of index structures have been developed for ANNS. Tree-based methods recursively partition the vector space using hierarchical structures but suffer from the "curse of dimensionality" as dimensionality increases. Hashing-based methods map high-dimensional vectors to compact binary codes using locality-sensitive hashing (LSH) [32], enabling fast similarity estimation through Hamming distance. Graph-based methods construct a proximity graph where nodes represent database items and edges connect similar items. During retrieval, a greedy traversal starting from entry points iteratively expands the candidate set by examining neighbors of visited nodes. The Hierarchical Navigable Small World (HNSW) algorithm [28] builds a multi-layered graph where upper layers provide sparse long-range connections for coarse navigation and lower

layers provide dense local connections for fine-grained search. Graph-based methods have demonstrated superior performance on high-dimensional data and are widely adopted in production systems [33].

A key property for efficient graph traversal is the **triangle inequality**: for any three points a, b, c in a metric space, $d(a, c) \leq d(a, b) + d(b, c)$. When this property holds, greedy graph traversal is guaranteed to follow optimal paths. However, learned neural relevance metrics often violate the triangle inequality [29], forcing the exploration of redundant edges and degrading retrieval efficiency.

Modern large-scale video retrieval and recommendation systems adopt a retrieval-and-ranking architecture [7, 8]. The retrieval stage reduces a billion-scale candidate corpus to a thousand-scale candidate set using lightweight models, which then serves as input for more sophisticated ranking models. The two-tower architecture [34] is widely adopted for the retrieval stage: it encodes queries and items through separate neural networks into embeddings compared via efficient similarity metrics such as inner product or cosine similarity. This architecture enables pre-computation and indexing of item representations, allowing real-time retrieval through ANNS. Beyond simple distance metrics, recent works use DNN-based learnable metrics [35, 36] to estimate relevance through high-dimensional computation over embeddings, capturing more complex interaction patterns than hand-designed similarity functions [29, 37].

The ANNS infrastructure and retrieval-and-ranking paradigm described above are shared across video retrieval and large-scale recommendation systems: in both settings, embeddings are pre-computed and indexed, and queries are answered through graph-based approximate search. In video recommendation, the retrieval stage instantiates the same infrastructure problem faced by deployment-scale video retrieval: learned relevance signals must be searched efficiently over massive candidate collections. The **search scale** of video retrieval therefore concerns whether learned retrieval signals remain searchable

and executable within such large-scale serving infrastructure. The challenges of designing search-compatible relevance metrics, efficient graph traversal, and GPU-accelerated execution under high concurrency are common to both domains.

2.2 Literature Review

This section surveys existing work along three research themes that span the contributions of this thesis: the evolution of video embedding models and their training paradigms (§2.2.1), the development of video retrieval benchmarks and evaluation protocols (§2.2.2), and the integration of learned retrieval metrics with large-scale graph-based search systems (§2.2.3). These three themes correspond to the semantic, observational, and search scaling challenges introduced in Chapter 1, and together trace the end-to-end pipeline from representation learning through evaluation to deployment-scale retrieval.

2.2.1 Video Embedding Models

The development of video embedding models can be traced through two major phases. The first phase extended CLIP [2] to video by introducing temporal modeling: CLIP4Clip [3] pioneered temporal aggregation over frame-level features, followed by InternVideo2 [13], X-CLIP [38], VideoCLIP-XL [39], and LanguageBind [40], which developed more sophisticated spatiotemporal encoders. These models, all built on the dual-encoder paradigm, achieved strong results on standard text-to-video benchmarks but were inherently constrained by their modality-specific architecture: the text encoder cannot process visual inputs, and the video encoder cannot process textual queries, limiting their applicability to composed or visual retrieval tasks.

The second phase shifted to MLLM-based backbones that process multiple modalities within a unified architecture. Models such as LLaVE [41], UNITE [4], and VLM2Vec-

V2 [42] demonstrated that leveraging the language understanding and visual generalization of large pretrained models could yield strong embedding performance. Subsequent models including GME [12], BGE-VL [43], UniME [44], and B3 [45] further advanced the MLLM-based paradigm. Qwen3-VL-Embedding [46] showed the potential of large-scale supervised training for MLLM-based embeddings. Despite these advances, a systematic evaluation of such models across diverse retrieval tasks and semantic domains has been lacking, leaving their true generalization capabilities unclear.

A further challenge lies in the heterogeneity of multi-task supervision. When training data spans diverse query formats (textual, composed, visual) and semantic domains (coarse-grained, fine-grained, partially relevant, long-context, long-video), naively mixing them can lead to gradient interference: gradient signals from complex tasks may be submerged by simpler ones, leading to suboptimal convergence. Curriculum learning [47], which schedules training examples from easier to harder tasks, offers a principled approach to such optimization challenges, but its application to heterogeneous multi-task training for video retrieval embeddings has not been systematically explored. This gap motivates the curriculum-based training methodology proposed in Chapter 3.

2.2.2 Video Retrieval Benchmarks and Evaluation

Text-to-video retrieval has progressed from matching coarse phrases to parsing fine-grained spatio-temporal descriptions. Early datasets established the foundational paradigm: MSR-VTT [6] introduced large-scale video-caption pairing with short clips, while DiDeMo [18] focused on temporal moment localization.

Subsequent benchmarks extended the scope in complementary directions. In the fine-grained dimension, CaReBench [9] incorporates spatio-temporal grounding annotations, UltraVideo [48] targets scene understanding, and CameraBench [49] evaluates camera motion comprehension. In the long-video dimension, LoVR [11] targets retrieval over

hour-long videos, while TVR [50] and ActivityNet Captions [51] provide fine-grained temporal annotations for partially relevant video retrieval [52]. Beyond text-to-video retrieval, new paradigms have emerged, including composed queries with text-image pairs (e.g., CoVR [10, 53], MomentSeeker [54]) and purely visual queries in egocentric contexts [55]. Additional benchmarks such as VDC [15] provide multi-granular caption annotations, and PE-Video [56] and DREAM-1K [57] offer keyword-based and event-based retrieval annotations, respectively.

Despite this progress, evaluation remains fragmented. Each benchmark addresses only one or two dimensions of the retrieval problem, covering either query modality or semantic granularity but rarely both. Models are optimized and tested on isolated benchmarks, preventing a holistic understanding of their generalization across heterogeneous retrieval settings. No existing evaluation setting provides the combination of long-form visual evidence and multi-granular queries required to study how retrieval performance scales with query specificity across diverse semantic conditions.

These gaps motivate two complementary efforts in this thesis. Chapter 3 introduces a unified evaluation protocol across 16 existing datasets, establishing a standardized diagnostic framework that exposes systemic capability deficiencies across current models. Chapter 4 introduces MG-LVR, the first benchmark that combines long-form videos with multi-granular query categorization, enabling the first controlled study of how spatiotemporal sampling interacts with retrieval granularity. The scaling analysis built upon MG-LVR further reveals a training-inference density mismatch that motivates VEterfall, a progressive dense-to-sparse self-distillation framework [58] for efficient video embedding inference.

2.2.3 Large-Scale Retrieval and Approximate Nearest Neighbor Search

As defined in §2.1.4, graph-based ANNS methods such as HNSW [28] have become the dominant approach for high-dimensional similarity search, with subsequent works developing K-nearest neighbor graphs [59], relative neighborhood graphs [60], and GPU-optimized constructions [61]. These methods achieve sub-linear query time by constructing proximity graphs that enable greedy traversal toward the query, but their effectiveness depends critically on the properties of the similarity metric used during traversal. In parallel, the retrieval metrics used in practice have evolved from simple distance functions such as cosine similarity to learned DNN-based metrics that capture more complex relevance patterns (as discussed in §2.1.4).

A critical challenge arises at the intersection of these two trends: neural relevance metrics often violate the triangle inequality [29], a property that ensures optimal pathfinding in graph traversal (see §2.1.4). This violation forces the exploration of redundant edges and degrades retrieval efficiency, as the greedy traversal may follow paths that appear promising according to the learned metric but do not lead to the true nearest neighbors. Early industrial frameworks such as SL2G [37] first used DNN-based metrics for graph-based ANNS in real-world recommendations, demonstrating that learned metrics could improve retrieval accuracy despite their incompatibility with graph traversal. NANN [29] extended this by improving the DNN-based metric for better accuracy and adaptivity, but the fundamental tension between metric expressiveness and search compatibility remained unresolved.

Beyond metric design, the computational efficiency of large-scale retrieval under high concurrency poses additional challenges. In production systems serving hundreds of millions of users, retrieval requests arrive asynchronously, each triggering divergent search paths through the graph. The resulting irregular neural computations, where each path vis-

its a different set of candidate items requiring relevance evaluation, lead to frequent GPU kernel launches with fixed latency overhead that dominates runtime when individual computations are small [27, 31]. The co-design of learned retrieval metrics with graph traversal efficiency and GPU-parallel execution remains an open problem. This gap motivates the search-scale contributions proposed in Chapter 5, where the retrieval stage of video recommendation serves as the deployment context for methods that address the shared infrastructure challenges of large-scale embedding retrieval.

CHAPTER 3

TOWARDS UNIVERSAL VIDEO RETRIEVAL: GENERALIZING VIDEO EMBEDDING VIA SYNTHESIZED MULTIMODAL PYRAMID CURRICULUM

Chapter Abstract. The prevailing video retrieval evaluation suffers from fragmented benchmarks that obscure true model generalization. To diagnose this limitation, we conduct a structured assessment across 16 established public datasets, revealing critical capability gaps, particularly pronounced deficiencies in fine-grained, temporal, and partially relevant video retrieval. The discoveries directly motivate the holistic design of our framework: V-SynFlow synthesizes 1.55 million high-quality, multi-task video-text pairs explicitly targeting the uncovered deficiencies, while the Modality Pyramid curriculum trains the General Video Embedder (GVE) by leveraging inherent task hierarchies for progressive learning. GVE achieves state-of-the-art zero-shot performance across diverse scenarios. Our analysis further demonstrates that conventional benchmarks poorly correlate with holistic retrieval capability, and identifies partially relevant retrieval as a pivotal yet overlooked real-world challenge. This work establishes a practical pathway toward universal video retrieval by tightly coupling evaluation diagnostics with structured training on diverse synthesized data.

3.1 Introduction

Video retrieval is a critical and challenging task for modern search engines and recommendation systems, requiring effective video embedding models [1]. Early efforts extended Contrastive Language-Image Pretraining (CLIP) [2] to video [62]. Currently, Multimodal Large Language Models (MLLMs) are rapidly displacing CLIP due to the superior language

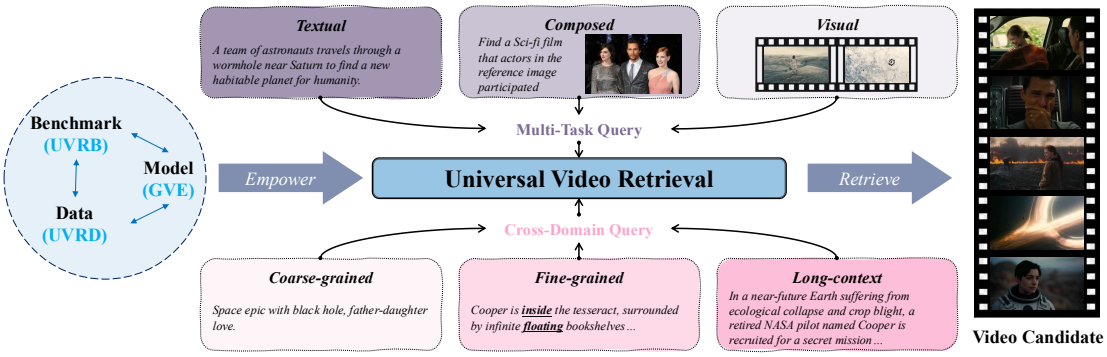


Figure 3.1: Universal Video Retrieval organizes heterogeneous retrieval settings into a unified capability space.

understanding and visual generalization capabilities of these models [4]. Current practices involve training these models on massive datasets with simple and noisy text annotations (e.g., WebVid [5]), which yields strong results for coarse-grained text-to-video retrieval on standard benchmarks (e.g., MSRVTT [6]).

However, these models struggle with the complexity of diverse real-world video retrieval scenarios (Figure 3.1). First, a narrow semantic distribution renders these models less effective in fine-grained queries required to understand spatial relations or temporal dynamics [9], as well as in long-context retrieval within lengthy videos [11]. Second, the scope of applicable tasks is restricted, with limited support for diverse query formats, such as composed retrieval using text-and-image pairs. Furthermore, these capability gaps are often obscured by the fragmented evaluation landscape, where models are optimized and tested on isolated benchmarks, preventing a clear understanding of comprehensive generalization.

Therefore, to establish a framework for universal video retrieval that supports multi-domain, multi-granularity, and multi-task capabilities, *two critical hurdles in data and modeling* must be addressed: (1) *Large-Scale Quality-Controlled Data Synthesis*: Existing datasets are either too small or are biased, while collecting a new, massive dataset is prohibitively expensive. Growing works attempt to synthesize data to address this challenge [10, 14, 15]. However, these resources often exhibit uneven quality and distri-

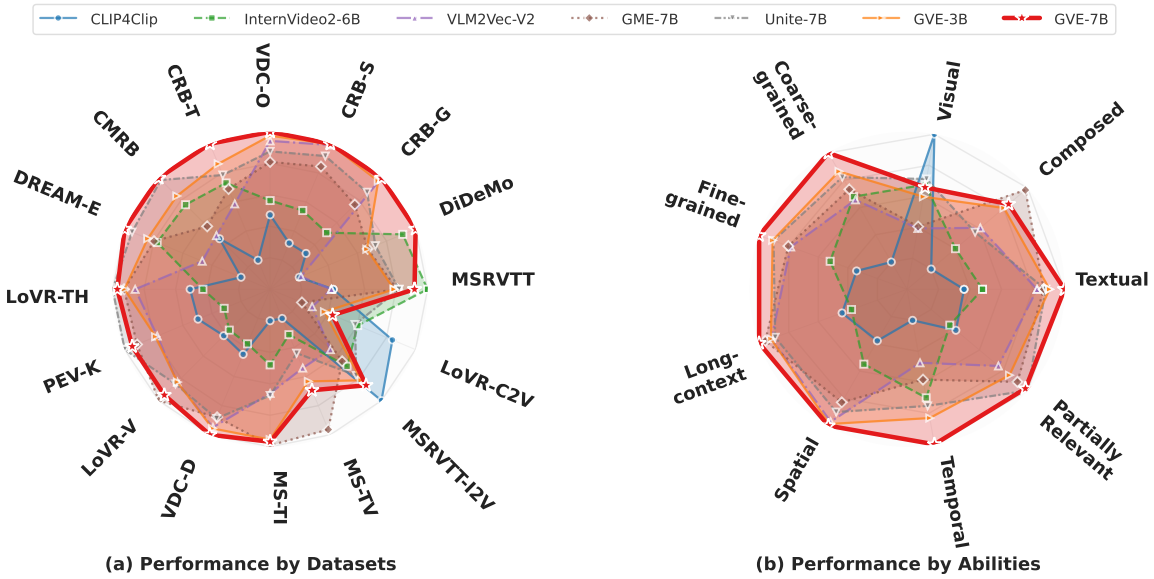


Figure 3.2: Model performance on UVRB for 16 datasets and 9 abilities (3 main tasks and 6 (sub-) domains).

bution. Therefore, gaining precise control over the properties of large-scale, cross-domain, multi-task data via a unified synthesis process is a primary challenge. (2) *Interconnected Multi-task Representation Learning*: A critical, often-ignored aspect is the inherent hierarchy among tasks. Foundational abilities such as spatial perception (e.g., object recognition) serve as building blocks for higher-order temporal reasoning (e.g., action recognition). This principle is evidenced by the remarkable success of image-only trained models like GME [12] on video retrieval tasks (see Figure 3.2), highlighting latent cross-task adaptability. Nevertheless, conventional models that assume task independence fail to capitalize on this structure, which is a key to unlocking superior generalization.

To this end, we propose a holistic framework that couples targeted data synthesis with stable modeling. First, to expose the concrete limitations of current approaches, we perform a standardized evaluation across 16 established public datasets, assessing 14 state-of-the-art models under unified preprocessing and metrics. Second, directly guided by the capability gaps identified in this evaluation (e.g., deficiencies in temporal and composed retrieval), we design **V-SynFlow**. This workflow transforms raw videos into a targeted,

high-quality dataset, the **Universal Video Retrieval Dataset (UVRD)**, consisting of over 1.55 million multi-task pairs. Third, to ensure stability during heterogeneous training, we apply the **Modality Pyramid**, a training curriculum that leverages inherent task dependencies. It optimizes the **General Video Embedder (GVE)** by prioritizing foundational tasks before progressing to complex ones, facilitating smooth gradient convergence.

Extensive experiments validate the effectiveness of the GVE model (Figure 3.2). In addition to strong zero-shot performance, the analysis reveals that performance on conventional benchmarks correlates poorly with overall retrieval capability; meanwhile, partially relevant video retrieval can serve as a representative indicator of universal video retrieval.

The main contributions of this work are summarized as follows: (1) **Data**: We propose V-SynFlow, a synthesis pipeline that produces UVRD with over 1.55 million video-text pairs to explicitly address the data deficiencies identified in current models. (2) **Training**: We introduce the Modality Pyramid curriculum, a training strategy that mitigates modality interference in heterogeneous multi-task learning, enabling the GVE model to achieve stable and strong zero-shot generalization. (3) **Analysis**: Rather than proposing a novel benchmark, we establish a unified evaluation protocol across 16 existing datasets. This standardized assessment systematically diagnoses model universality and offers new empirical insights beyond traditional evaluation paradigms.

3.2 Related Works

Video Retrieval. Text-to-video retrieval has progressed from matching coarse phrases to parsing fine-grained spatio-temporal descriptions [6]. While recent benchmarks have advanced beyond simple recognition by incorporating detailed annotations like spatio-temporal grounding (CaReBench [9]), scene understanding (UltraVideo [48]), and camera motion (CameraBench [49]), they remain specialized. Concurrently, the scope of retrieval

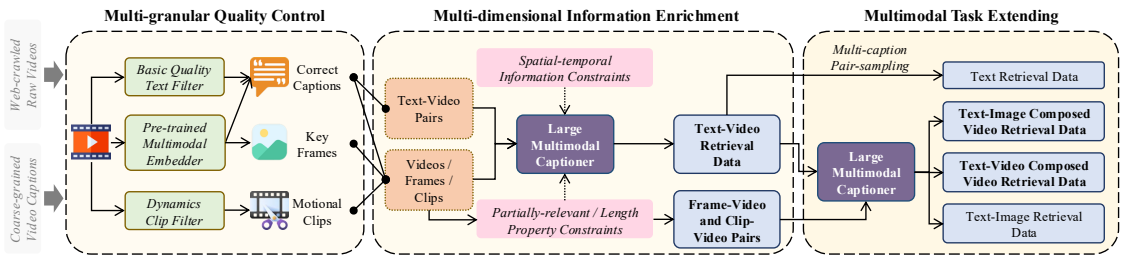


Figure 3.3: V-SynFlow: a multi-stage synthesis workflow for diverse video retrieval data.

has expanded to include new paradigms like composed queries with text-image pairs (e.g., CoVR [10, 53]) and purely visual queries in egocentric contexts [55]. Despite these advances, the field remains fragmented, with models and evaluation siloed within specific tasks or domains. This prevents a holistic understanding of a model's true generalization capabilities.

Video Embedding Models. Video embedding models have evolved from the adaptation of image-centric ones, such as CLIP [2], to powerful, larger language-based encoders. Early methods such as CLIP4Clip [3] and InternVideo2 [13] added temporal modules to CLIP but inherited its limitations in complex language understanding and long-context processing [63, 64]. To overcome these issues, recent work leverages Multimodal Large Language Models (MLLMs) as video embedders. Models like LLaVE [41], UNITE [4], and VLM2Vec-V2 [42] achieve strong performance on benchmarks by training on text-image and text-video data. However, this data has so far been limited for learning generalizable embeddings, and these video embedding models have not been systematically evaluated on more complex video retrieval tasks and across diverse video domains.

3.3 Preliminary

To bridge the persistent gap of fragmented evaluation protocols, we formulate the core problem from a holistic perspective by introducing the unified paradigm of Universal Video Retrieval (UVR). This preliminary setup provides an objective diagnostic foundation to motivate our method contributions.

3.3.1 Formulations

Definition 3.3.1 (Universal Video Retrieval (UVR)) *Given a matching pair comprising a query q and a video v, UVR aims to learn a θ-parameterized model $E_\theta(\cdot)$ to compute a relevance score $s_{q\to v} = \cos(E_\theta(q), E_\theta(v))$. This score must be consistently higher than the similarity scores of irrelevant pairs across any query format or data domain. Specifically, this paradigm categorizes the retrieval space into two primary dimensions:* ***(1) Query Format*** *incorporates Textual (TXT, e.g., natural language), Composed (CMP, combinations such as text and image or text and video), and Visual (VIS, pure image or video queries).* ***(2) Data Domain*** *incorporates Coarse-grained (CG, high-level semantics), Fine-grained categories including Spatial (S, object), Temporal (T, dynamics), and Partially Relevant (PR, partial information), alongside Long-context (LC, lengthy inputs).*

This formalization integrates isolated retrieval tasks into a comprehensive taxonomy, delineating the full spectrum of capabilities required for a truly universal video retrieval system.

3.3.2 Unified Evaluation Protocol

To operationalize the UVR paradigm and systematically diagnose generalization bottlenecks, we establish a standardized evaluation protocol across 16 public video retrieval

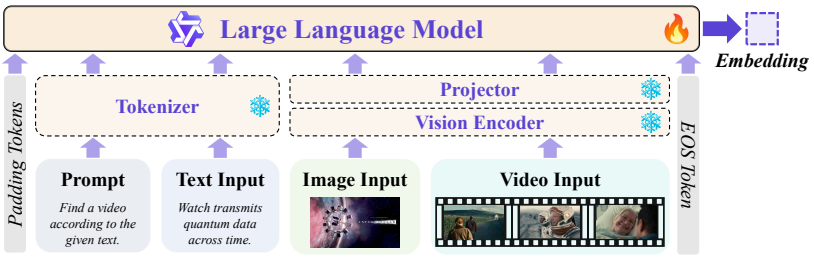


Figure 3.4: The architecture of GVE, an MLLM-based embedding model. We only fine-tune the LLM part. GVE inputs compositional multimodal elements and outputs a high-dimensional vector as an embedding. Details are in Appendix A.1.6.

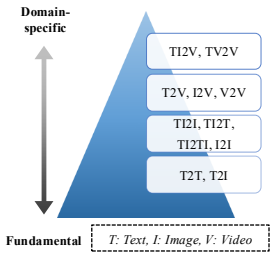


Figure 3.5: Modality Pyramid: simpler tasks lay the foundation for specific ones.

datasets. The consolidation of these existing resources into a unified evaluation suite—which we refer to as the Universal Video Retrieval Benchmark (UVRB) protocol—provides a comprehensive framework for assessing model universality against the exhaustive query formats and domain complexities outlined in Definition 3.3.1.

To guarantee an objective assessment, all datasets undergo identical preprocessing, including uniform frame sampling, query normalization per task type, and strict adherence to a vision-only input scope. The evaluation follows a strictly model-agnostic protocol. Embeddings are extracted directly from the frozen representation space, and relevance is computed solely via cosine similarity without task-specific post-processing or learnable projections. This standardization eliminates heuristics and isolates the visual-semantic alignment capabilities of the backbones.

This comprehensive diagnostic framework (detailed in Appendix A.1.1) exposes systemic capability deficiencies across current models, directly motivating the necessity and design of our targeted data synthesis pipeline and heterogeneous task curriculum in the subsequent methodology.

3.4 Methodology

This section introduces a co-designed framework for universal video retrieval. Guided by UVR, we first propose V-SynFlow. This workflow synthesizes the Universal Video Retrieval Dataset (UVRD), comprising 1.55 million targeted video-text pairs to explicitly supplement general retrieval data. Subsequently, we introduce the Modality Pyramid curriculum, a strategy designed to stabilize the optimization of the General Video Embedder (GVE) by leveraging inherent task hierarchies across heterogeneous modalities.

3.4.1 Scalable Synthesis of Cross-Domain Multi-Task Video Retrieval Data

Training a universal video embedder is fundamentally impeded by a deficiency of high-quality supervision for divergent tasks. Guided by the concrete capability gaps identified in the preliminary diagnostics, such as severe shortcomings in fine-grained temporal reasoning and composed retrieval, we introduce **V-SynFlow** (Figure 3.3). This pipeline transforms weakly annotated web videos from raw datasets (e.g., PVD [56], InternVid-FLT [65], and WebVid [5]) into structured, high-fidelity, and gap-targeted multi-task training instances. Through this targeted approach, we construct the **Universal Video Retrieval Dataset (UVRD)**, containing over 1.55 million descriptive pairs tailored for multidimensional training enhancement (detailed in Appendix A.1.3). Specifically, the V-SynFlow workflow proceeds in three stages as follows. Detailed procedures and the applied prompts for synthesis are provided in Appendix A.1.5.

Multi-granular Quality Control. Given a raw corpus $\mathcal{D} = \{(v_i, t_i)\}$, we produce a high-fidelity asset pool, $\mathcal{A}_{tfc}$. The process applies a filter cascade: *Annotation Rectification* removes non-descriptive text; *Cross-Modal Consistency Filtering* discards pairs where the similarity measured by a pretrained embedder $\Phi(\cdot)$ (e.g., GME-7B [12]) falls below a prede-

fined threshold (empirically set to 0.25, with detailed ablation provided in Appendix A.1.7); and *Temporal Dynamics Filtering* forcibly removes static content to ensure sufficient temporal signals. The resulting asset pool $\mathcal{A}_{tfc}$ contains a set of validated videos $\{v_j\}$, the original captions $\{t_j\}$, and corresponding sets of extracted frames $\{f_{jk}\}$ alongside cropped clips $\{c_{jl}\}$.

Multi-dimensional Information Enrichment. We leverage the filtered assets within $\mathcal{A}_{tfc}$ to generate richer data structures. To create an enriched text-video dataset $\mathcal{D}^{+}$, we utilize an MLLM, $\mathcal{M}_{\text{cap}}$ (e.g., Keye-VL-8B [66]), as a conditional captioning engine. For each video v_j, the engine synthesizes multiple captions $\{t'_{jk}\}$ conditioned by diagnostic-driven information profiles. Specifically, because the diagnostic evaluation revealed a pronounced spatial-temporal decoupling wherein temporal skills constitute the primary bottleneck, we deliberately skew the synthesis distribution: 60% temporal, 30% spatial, and 10% targeted at partially relevant or abstract semantics, informed by our preliminary diagnostic results on UVRB. By sampling one caption for each video, we obtain a set of novel high-quality text-video pairs $\{(v_j, t'_{jk})\}$. Furthermore, we pair videos with the corresponding visual components to construct collections of visual pairs $\mathcal{P}_{f\leftrightarrow v}$ and $\mathcal{P}_{c\leftrightarrow v}$, resulting in frame-to-video $\{(f_{jk}, v_j)\}$ and clip-to-video $\{(c_{jl}, v_j)\}$ instances. To mitigate potential hallucination biases inherent in generative MLLMs, we enforce strict instructional prompts and rely on the aforementioned cross-modal consistency filtering to automatically discard misaligned generations.

Multimodal Task Extension. The final stage assembles the unified training corpus, $\mathcal{D}_{\star}$. To explicitly address the composed retrieval deficiency, we synthesize complex composed tasks by leveraging $\mathcal{P}_{f\leftrightarrow v}$ and $\mathcal{P}_{c\leftrightarrow v}$. For each visual pair, such as (f_{jk}, v_j), the engine $\mathcal{M}_{\text{cap}}$ generates a query text $t^{f_{jk}\to v_j}$ describing the temporal evolution, forming a training instance $((t^{f_{jk}\to v_j}, f_{jk}), v_j)$. Additionally, basic alignment tasks (e.g., text-image pairs) are sampled

from $\mathcal{A}_{tfc}$. For each video, two candidates from the unselected synthesized captions are mapped as text-to-text pairs. The resulting dataset $\mathcal{D}_\star$ provides the comprehensive mixture of tasks essential for training a universal embedder.

Human Validation

To assess the quality of the automatically generated UVRD, human expert evaluation was conducted on a random sample of 100 instances. Validation confirmed a 95% factual accuracy rate. While this human-verified sample is a preliminary subset, it corroborates the efficacy of our automated multi-granular quality control pipeline, which serves as the primary scalable safeguard for the entire 1.55 million-pair corpus. Detailed analysis of the minor erroneous instances and the corresponding error taxonomy are in Appendix A.1.12.

3.4.2 Modality Pyramid: Customized Curriculum Contrastive Learning

The synthesized dataset provides a rich mixture of retrieval tasks spanning diverse query formats and data domains. To embed these multimodal inputs into a unified space, we introduce the **General Video Embedder (GVE)** (Figure 3.4), a multimodal encoder derived from Qwen2.5-VL [67] to inherit the pretrained vision-language alignment of the backbone model. For arbitrary modality combinations, the GVE model fuses the tokenized prompt and text inputs with projected visual features into a joint input sequence. The LLM processes this sequence to autoregressively produce representations. We extract the final embedding via last token pooling and ℓ_2-normalization for retrieval.

However, naively mixing this highly heterogeneous data leads to severe modality interference and optimization instability. In early training stages under random scheduling, gradient signals from complex temporal and composed tasks are frequently submerged by sim-

Table 3.1: Video retrieval performance for datasets of UVRB. The AVG values are averaged over 16 datasets. For each column: highest score is **bolded**, second-highest is underlined. Metrics: R@1 (Recall@1), R@10 (Recall@10), P@1 (Precision@1).

Model	AVG	MSRVTT R@1	DiDeMo R@1	CRB-G R@1	CRB-S R@1	VDC-O R@1	CRB-T R@1	CMRB R@10	DREAM-E R@1
CLIP4Clip	0.390	0.333	0.297	0.511	0.497	0.620	0.289	0.280	0.191
ViCLIP	0.352	0.386	0.306	0.447	0.437	0.530	0.349	0.229	0.235
VideoCLIP-XL	0.491	0.443	0.403	0.828	0.839	0.735	0.487	0.274	0.263
LanguageBind	0.487	0.479	0.421	0.716	0.687	0.759	0.466	0.290	0.280
InternVideo2-1B	0.404	0.449	0.404	0.586	0.568	0.644	0.470	0.355	0.242
InternVideo2-6B	0.427	**0.485**	0.418	0.608	0.612	0.650	0.455	0.346	0.271
GME-2B	0.488	0.390	0.303	0.690	0.718	0.715	0.400	0.298	0.240
Unite-2B	0.480	0.367	0.298	0.699	0.723	0.727	0.409	0.284	0.223
VLM2Vec-V2	0.508	0.330	0.299	0.828	0.843	0.775	0.410	0.286	0.228
BGE-VL	0.443	0.337	0.318	0.690	0.688	0.639	0.359	0.225	0.212
UniME-7B	0.521	0.351	0.335	0.815	0.827	0.743	0.476	0.317	0.293
B3-7B	0.511	0.282	0.350	0.815	0.825	0.768	0.415	0.312	0.216
GME-7B	0.530	0.436	0.377	0.740	0.767	0.731	0.442	0.304	0.274
Unite-7B	0.538	0.439	0.386	0.798	0.804	0.753	0.472	0.351	0.279
GVE-3B	0.544	0.431	0.376	0.850	0.846	0.786	0.496	0.363	0.280
GVE-7B	**0.573**	0.464	**0.433**	**0.865**	**0.847**	**0.794**	**0.539**	**0.398**	**0.302**

Model	LoVR-TH R@10	PEV-K R@1	LoVR-V R@1	VDC-D R@1	MS-TI P@1	MS-TV P@1	MSRVTT-I2V R@1	LoVR-C2V R@1
CLIP4Clip	0.338	0.179	0.360	0.566	0.173	0.183	**0.924**	0.503
ViCLIP	0.202	0.075	0.230	0.395	0.283	0.243	0.846	0.433
VideoCLIP-XL	0.439	0.229	0.380	0.820	0.230	0.223	0.861	0.403
LanguageBind	0.425	0.303	0.540	0.679	0.228	0.233	0.827	0.463
InternVideo2-1B	0.298	0.026	0.280	0.485	0.265	0.230	0.794	0.368
InternVideo2-6B	0.302	0.086	0.330	0.516	0.235	0.205	0.868	0.452
GME-2B	0.446	0.354	0.530	0.839	**0.350**	**0.340**	0.827	0.366
Unite-2B	0.445	0.355	0.570	0.792	0.250	0.233	0.863	0.445
VLM2Vec-V2	0.492	0.324	0.610	0.913	0.275	0.250	0.841	0.385
BGE-VL	0.387	0.184	0.550	0.722	0.303	0.233	0.779	0.465
UniME-7B	0.504	0.323	0.480	0.847	0.310	0.305	0.867	**0.537**
B3-7B	0.462	0.387	0.590	0.853	0.275	0.265	0.884	0.471
GME-7B	0.523	0.396	**0.710**	0.865	0.348	0.333	0.860	0.370
Unite-7B	**0.555**	**0.440**	0.620	0.871	0.278	0.230	0.883	0.448
GVE-3B	0.522	0.330	0.610	0.918	0.340	0.268	0.891	0.403
GVE-7B	0.542	0.413	0.680	**0.948**	0.343	0.280	0.899	0.415

pler spatial alignment tasks, leading to erratic loss convergence and suboptimal performance on complex tasks. To ensure stable optimization across heterogeneous formats, we propose **Modality Pyramid** (Figure 3.5), a curriculum that schedules training from atomic to composite tasks for progressive knowledge acquisition. This bottom-up design is grounded in empirical evidence: after a single epoch, a clear task hierarchy emerges based on alignment scores, descending from text-image (0.65) down to text-video (0.46) and composed video (0.43) tasks (Appendix A.1.13). The curriculum functions as a stabilizing strategy, guiding the embedding model to master perceptual primitives prior to advancing to complex integration, thereby mitigating gradient submergence.

Alignment-aware Dynamic Scheduling. At the beginning of each epoch t, we estimate the alignment of every task $k \in \mathcal{K}$ using a *prober model* Ψ_t. For $t = 1$, Ψ_1 is a robust off-the-shelf embedder (e.g., GME-7B [12]). The selection of a robust, MLLM-based prober is deliberate to ground the curriculum in reliable and semantically rich task difficulty estimates for a stable initial anchor. For $t > 1$, Ψ_t utilizes the GVE checkpoint from the conclusion of epoch $t - 1$. To prevent computational bottlenecks, this probing evaluation is performed offline using a randomly downsampled subset. For each task, we sample $N_p = 1000$ positive pairs and compute the relevance score as the average cosine similarity: $R_k^{(t)} = \frac{1}{N_p} \sum_i \cos(\Psi_t(x_i), \Psi_t(y_i))$. A higher $R_k^{(t)}$ indicates better current alignment. During epoch t, tasks are sampled with a probability $P^{(t)}(k) \propto \exp(R_k^{(t)}/\sigma(t))$, where the temperature $\sigma(t)$ increases linearly from σ_{min} to σ_{max} ($\sigma_{\text{min}} = 0.1$ and $\sigma_{\text{max}} = 1.0$ by default) across the total duration of the training epochs. This topological progression ensures an initial focus on well-aligned tasks while progressively encouraging challenging exploration.

Unified Contrastive Optimization. The GVE model is trained using a symmetric InfoNCE loss [68] across all scheduled tasks. To strengthen discrimination, we augment in-batch negatives with hard negatives mined from a large external corpus utilizing the identical prober Ψ_t. To mitigate the risk of false negatives (i.e., visually similar but distinct videos that happen to share target semantics), we apply a strict similarity margin constraint during mining, discarding any candidates that exceed a similarity threshold of 0.90 with the query. For a query–target pair (q, v^+), the loss contrasts v^+ against both other positive samples in the batch and the top-K retrieved hard negatives. The similarity is computed as $s_{q \to v} = \cos(E_\theta(q), E_\theta(v))$, and the final loss is symmetrized over the query and target

Table 3.2: Video retrieval performance by specific abilities (tasks and domains) on UVRB. The AVG values are averaged over tasks (textual (TXT), composed (CMP), visual (VIS)) and domains (coarse-grained (CG), fine-grained (FG), long-context (LC)) video retrieval tasks. Besides, we provide sub-domain results, including spatial (S), temporal (T), partially relevant (PR). For each column: highest score is **bolded**, second-highest is underlined.

		Tasks			Domains			Sub-domains		
Model	**AVG**	**TXT**	**CMP**	**VIS**	**CG**	**FG**	**LC**	**S**	**T**	**PR**
CLIP4Clip	0.416	0.401	0.178	**0.714**	0.380	0.360	0.463	0.559	0.285	0.236
ViCLIP	0.375	0.336	0.263	0.640	0.380	0.315	0.313	0.484	0.289	0.171
VideoCLIP-XL	0.510	0.550	0.227	0.632	0.558	0.493	0.600	0.787	0.381	0.310
LanguageBind	0.508	0.543	0.231	0.645	0.539	0.479	0.610	0.723	0.378	0.336
InternVideo2-1B	0.420	0.422	0.248	0.581	0.480	0.403	0.383	0.606	0.413	0.189
InternVideo2-6B	0.445	0.448	0.220	0.660	0.504	0.417	0.423	0.631	0.400	0.220
GME-2B	0.416	0.539	**0.345**	0.597	0.461	0.471	0.685	0.716	0.349	0.347
Unite-2B	0.507	0.536	0.242	0.654	0.455	0.471	0.681	0.725	0.347	0.341
VLM2Vec-V2	0.538	0.587	0.263	0.613	0.498	0.502	0.762	0.809	0.348	0.348
BGE-VL	0.480	0.497	0.268	0.622	0.448	0.406	0.636	0.664	0.292	0.261
UniME-7B	0.542	0.561	0.308	0.702	0.500	0.518	0.664	0.785	0.396	0.373
B3-7B	0.538	0.570	0.270	0.678	0.482	0.505	0.722	0.797	0.364	0.355
GME-7B	0.562	0.604	0.341	0.615	0.518	0.507	0.788	0.749	0.373	0.398
Unite-7B	0.559	0.609	0.254	0.666	0.541	0.539	0.746	0.779	0.412	**0.425**
GVE-3B	0.571	0.619	0.304	0.647	0.552	0.541	0.764	0.816	0.430	0.377
GVE-7B	**0.600**	**0.657**	0.312	0.657	**0.587**	**0.570**	**0.814**	**0.821**	**0.469**	0.419

directions:

$$\mathcal{L}_i^{(q\to y)} = -\log\left\{ \exp(s_{q_i\to y_i^+}/\tau_l) \Big/ \left[\exp(s_{q_i\to y_i^+}/\tau_l) + \sum_{j\neq i} \exp(s_{q_i\to y_j^+}/\tau_l) + \sum_{y_k^-\in\mathcal{H}_i} \exp(s_{q_i\to y_k^-}/\tau_l) \right] \right\} \tag{3.1}$$

where $s_{q\to v} = \cos(E_\theta(q), E_\theta(v))$ (with q and y denoting the query and target modalities, respectively), and τ_l represents a pre-defined temperature. The total loss constitutes the symmetric sum $\mathcal{L}_i = \frac{1}{2}(\mathcal{L}_i^{(q\to y)} + \mathcal{L}_i^{(y\to q)})$.

3.5 Experiments

3.5.1 Experimental Setups

Baselines. Our evaluation benchmarks 14 prominent baselines. Our selection spans a wide range of architectures, parameter sizes (from 87M to 8.3B), and training data

compositions. The models are broadly divided into two categories: First, traditional *CLIP-based* embedding models include CLIP4Clip [3], ViCLIP [65], VideoCLIP-XL [39], LanguageBind [40], and the InternVideo2 series (1B and 6B) [13]. Second, a more recent category of *MLLM-based* embedding models includes GME-2B [12], Unite-2B [4], VLM2Vec-V2 [42], BGE-VL [43], UniME-7B [44], B3-7B [45], GME-7B [12], and Unite-7B [4]. While we prioritize evaluating the zero-shot generalization capabilities of these models (strictly for our GVE series), we acknowledge that baselines extensively trained on in-domain data (e.g., Qwen3-VL-Embedding [46]) establish a supervised upper bound. A detailed discussion and comparison with these in-domain models is deferred to Appendix A.1.8.

Metrics. Our primary metric is Recall@1 (R@1), which measures whether the most relevant item is the correct one. For more challenging datasets with fuzzy queries (e.g., CMRB and LoVR-TH), we choose to report Recall@10 (R@10). Additionally, we use Precision@1 (P@1) for the MS-TI and MS-TV with multiple positive candidates.

Evaluation Implementations. All models are evaluated under the aforementioned unified protocol (UVRB) to ensure fair and reproducible comparisons across diverse domains. To isolate core representational power, we standardize the output pipeline: model parameters are loaded in bf16 precision, all embeddings are ℓ_2-normalized, and cosine similarity serves as the universal metric, with no model-specific post-processing or re-ranking. Critically, we follow [9] to remove any learnable projection heads (e.g., the MLP head in the InternVideo2 series), using the hidden state from the final backbone layer as the general-purpose embedding. On the input side, the evaluation is vision-only, with each video uniformly sampled into exactly 8 frames. To balance fairness with architectural diversity, CLIP-based models process frames at a fixed 224×224 resolution, while MLLM-based models are constrained to a budget of < 200 visual tokens per frame via adaptive resizing.

Table 3.3: Ablation study for UVRD and Modality Pyramid. **AVG of D**: average across datasets, **AVG of A**: average across abilities. For each column of each size of model (3B or 7B): highest score is **bolded**, second-highest is underlined.

	D	A	Tasks			Domains			Sub-domains		
Model	**AVG**	**AVG**	**TXT**	**CMP**	**VIS**	**CG**	**FG**	**LC**	**S**	**T**	**PR**
GVE-i-3B	0.528	0.558	**0.620**	0.237	0.632	0.532	0.521	**0.808**	0.781	0.402	**0.379**
GVE-s-3B	0.537	0.564	0.617	0.301	0.617	0.539	0.536	0.775	0.811	0.421	0.377
GVE-3B	**0.544**	**0.571**	0.619	**0.304**	**0.647**	**0.552**	**0.541**	0.764	**0.816**	**0.430**	0.377
GVE-i-7B	0.563	0.587	0.643	0.274	**0.678**	0.567	0.566	0.795	0.812	0.459	**0.426**
GVE-s-7B	0.568	0.594	0.648	**0.313**	0.662	0.576	0.563	0.804	0.814	0.458	0.418
GVE-7B	**0.573**	**0.600**	**0.657**	0.312	0.657	**0.587**	**0.570**	**0.814**	**0.821**	**0.469**	0.419

All input sequences are capped at 8192 tokens, and for models without native video support (e.g., BGE-VL), we implement a multi-image pipeline to ensure comparability. Given the substantial computational overhead of training 7B models over 13 million pairs, our main results are reported from a single definitive run. This aligns with standard evaluation protocols for large-scale MLLMs, where the massive scale of both training and zero-shot test sets naturally minimizes small-sample stochastic variance.

Others. Extended implementation details, including prompt templates, architecture specifics, and supplementary baseline comparisons, are comprehensively provided in the Appendix.

3.5.2 Main Results

Overall Performance. We evaluate GVE on the UVRB benchmark under a strictly zero-shot setting without any exposure to in-domain data during training. Although some competing models are exposed to training data corresponding to several test sets, our results in Table 3.1 and Table 3.2 show clear superiority and confirm the strong generalization of GVE. Specifically, GVE-7B achieves state-of-the-art results with mean scores of 0.573 across datasets and 0.600 across task categories. It outperforms UNITE-7B by $+6.5\%$ and $+7.3\%$ even though UNITE-7B may have seen in-domain data. GVE-7B leads in every major di-

mension including TXT at 0.657 versus 0.609, CMP at 0.312 versus 0.254, CG at 0.587, FG at 0.570, and LC at 0.814 versus 0.746. It also leads in fine-grained subdomains with S at 0.821 versus 0.779 and T at 0.469 versus 0.412. `UNITE-7B` shows strength in VIS and PR tasks but underperforms in compositional and temporal reasoning. Its performance is uneven and relies heavily on specific training data. The compact `GVE-3B` with 3.8B parameters scores 0.571, higher than `UNITE-7B` at 0.559 with over 7B parameters. This shows our gains come from better data synthesis and curriculum design not from model size or data leakage. Our smaller models match or beat larger ones under fair zero-shot evaluation. This advantage translates to top results while `UNITE-7B`'s strength on a few datasets reflects narrow capability. Furthermore, we evaluated GVE on video classification benchmarks in Appendix A.1.14, where our model demonstrates strong performance, confirming the embedding transferability. While the foundational backbone (Qwen2.5-VL) underwent large-scale pre-training, our GVE training strictly excludes any samples from the UVRB training and test sets, and the retrieval is performed in a model-agnostic, zero-shot manner without task-specific fine-tuning on the target domains.

Ablation Studies. To dissect the individual contributions of UVRD and Modality Pyramid, we conduct a series of ablation studies. First, we analyze their impact in Table 3.3. We define `GVE-I` as a baseline trained on public data with random task scheduling, and `GVE-S` as the same model enhanced with our synthesized UVRD. As the results show, integrating UVRD (`GVE-S` vs. `GVE-I`) yields a significant performance gain (+2.3% for GVE-3B). The benefit is particularly pronounced on complex tasks underrepresented in public data, with a remarkable +27% relative improvement on CMP. Subsequently applying the Modality Pyramid curriculum (`GVE` vs. `GVE-S`) further enhances performance across the board, confirming the benefit of structured learning.

Second, to specifically validate the hierarchical design of the curriculum, we compare

Table 3.4: Ablation on curriculum strategies at 3B scale.

Model Variant	Score
GVE (Ours)	**0.571**
GVE-s (Random)	0.564
GVE-r (Reverse)	0.545
GVE-ov (Video-Only)	0.550
GVE-nv (Non-Video)	0.521

our full model (GVE) against several variants with different training strategies according to their average scores across datasets, presented in Table 3.4. These variants include random scheduling (GVE-S), a reverse top-down curriculum (GVE-R), and models trained on only video (GVE-OV) or non-video (GVE-NV) data. The performance ranking is that GVE $>$ GVE-S $>$ GVE-OV $>$ GVE-R $>$ GVE-NV. The order offers two key insights. (1) *The learning sequence is critical:* Our structured curriculum (GVE) significantly outperforms both random (GVE-S) and reversed (GVE-R) scheduling. (2) *A synergy exists between modalities:* The full model (GVE) also surpasses training on uni-modal data sources alone (GVE-OV and GVE-NV). These findings confirm that a hierarchical integration of diverse data modalities is essential for optimal performance.

Data Scaling. Figure 3.6 shows the training-time scaling behavior that performance logarithmically improves with more data, but with diminishing returns. We quantify scaling efficiency by fitting a logarithmic model $y = a \ln x + b$ (x: data size, y: performance) and report the absolute and relative gain per $10\times$ data increase. On average across datasets (abilities), GVE-3B improves by $+7.4\%$ ($+7.1\%$) per decade, while GVE-7B gains $+5.4\%$ ($+5.4\%$). While GVE-3B exhibits higher scaling efficiency in relative terms, GVE-7B starts from a higher baseline, suggesting a trade-off between scaling slope and absolute capability. This result highlights the potential for scaling a more powerful video embedding model with training data at a larger scale. In addition, we also explore the test-time scaling of spatial

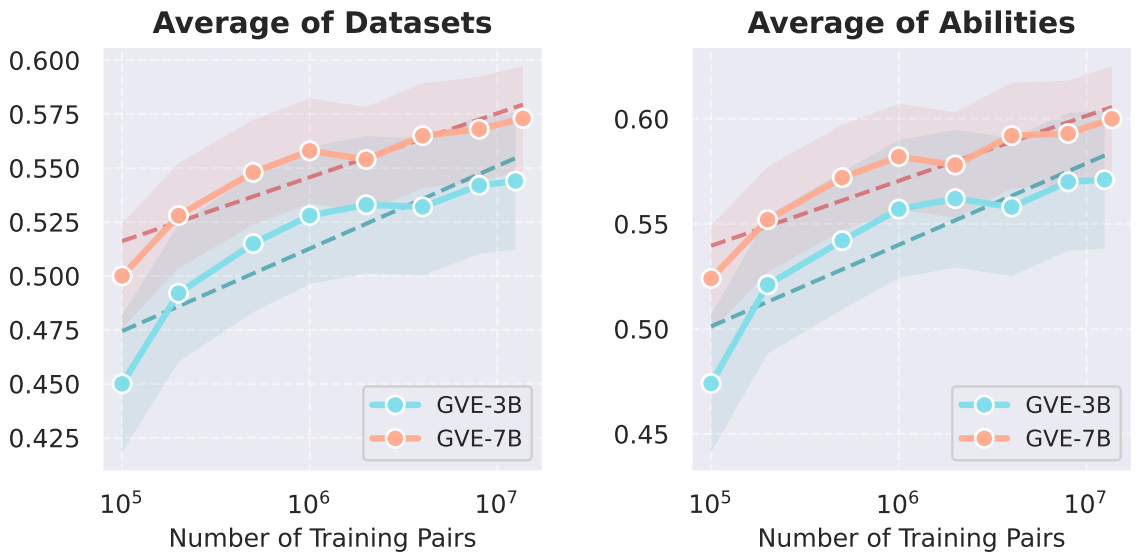


Figure 3.6: Performance effect from data scaling for GVE series. See Appendix A.1.10 for detailed results.

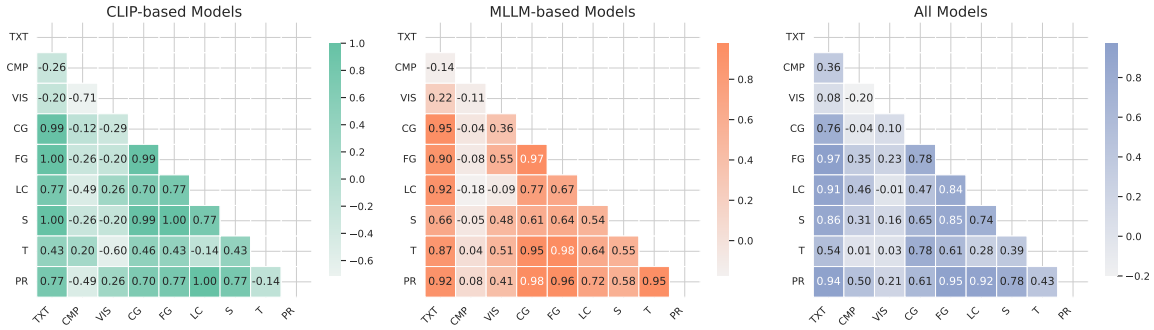


Figure 3.7: Correlation between dimensional abilities on UVRB for CLIP or MLLM-based models.

and temporal density (the frame number and resolution) in Appendix A.1.11 as a dimension of generalization.

3.5.3 Analysis of Dimensional Capabilities

Figures 3.7 and 3.8 reveal patterns in how video retrieval models develop capabilities and what their relationships are, as measured primarily by Pearson correlation (ρ) for their performance results. Here, we present several discoveries from the analysis in underexplored

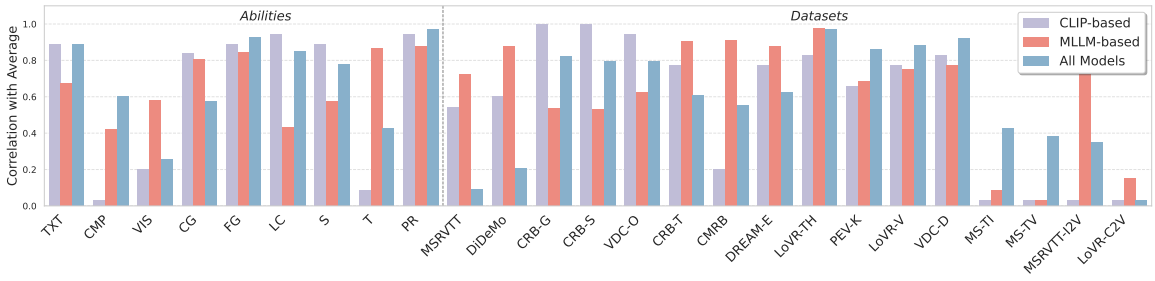


Figure 3.8: Correlation between averaged performance and abilities or datasets on UVRB.

perspectives.

Finding 1: Partially Relevant Video Retrieval Better Reflects Universality than Traditional Benchmarks. Standard benchmarks such as MSRVTT show a low correlation with average UVR performance ($\rho_{\text{avg}} = 0.58$), suggesting limited representativeness, likely due to overfitting or simplified task design. In contrast, fine-grained (FG), partially relevant (PR), and long-context (LC) retrieval exhibit strong mutual correlations ($\rho \geq 0.90$). More importantly, PR retrieval, although understudied, achieves the highest average correlation with overall performance ($\rho_{\text{avg}} = 0.97$), positioning it as the most reliable proxy for universal capability. We hypothesize this is because successful PR retrieval demands an embedding space that preserves a rich spectrum of information, e.g., from global context to subtle, localized details, rather than collapsing video semantics into a single "gist". The ability to maintain this multi-granular semantic richness is a core tenet of universal capability.

Finding 2: Disentangled Spatial and Temporal Representations. Models exhibit a marked decoupling between spatial (S) and temporal (T) representation ($\rho = 0.12$). This asymmetry is critical that temporal skills dominate fine-grained understanding ($\rho_{\text{T-FG}} = 0.98$), while spatial skills contribute minimally ($\rho_{\text{S-FG}} = 0.39$). This suggests current works fail to jointly model *when* and *where*, highlighting the requirements for inductive biases that

encourage spatiotemporal integration in video embedding.

Finding 3: Performance Divergence Between CLIP and MLLM-based Models. Model failures are architecture-dependent. For example, CLIP-based models are spatially biased ($\rho_{\text{S-CG}} = 0.99$) but temporally weak ($\rho_{\text{T-CG}} = 0.46$), leading to ability trade-offs. Compositional representation inversely correlates with visual accuracy ($\rho_{\text{CMP-VIS}} = -0.71$). And a near-zero link between temporal and long-context skills ($\rho_{\text{T-LC}} = -0.14$). In contrast, MLLM-based models demonstrate more balanced and integrated learning: superior semantic matching (PR-CG: MLLM $\rho = 0.98$ vs. CLIP 0.70) and stronger temporal-long-context coupling ($\rho_{\text{LC-T}} = 0.64$). Therefore, architecture fundamentally shapes the development of capability, and MLLM has been increasingly popular because of its generalizability in video embedding modeling.

Finding 4: Scaling Has Limited Impact on Visual Perception. Parameter scaling improves high-level semantic coherence, but yields negligible gains in low-level visual perception. Notably, the 87M-parameter `CLIP4Clip` (VIS: 0.714) defeats the 8B-parameter `UNITE-7B` (VIS: 0.702). Given the weak correlation between visual fidelity and overall retrieval success ($\rho_{\text{AVG-VIS}} = 0.26$), future progress requires targeted improvements in visual grounding.

3.6 Conclusion

This work advances the paradigm of universal video retrieval by shifting from isolated task optimization to a holistic, capability-driven framework. Guided by our standardized diagnostic protocol across existing benchmarks, we demonstrated that overcoming inherent model bottlenecks requires tightly coupling quality-controlled data synthesis with structured multi-task learning. Specifically, our V-SynFlow pipeline systematically bridges semantic

gaps via a million-pair UVRD dataset, while the Modality Pyramid curriculum stabilizes heterogeneous modality training for video embeddings. Consequently, our GVE achieves robust and leading generalization under zero-shot settings. Besides, our statistically rigorous evaluation provides the community with actionable empirical insights. To ensure full reproducibility, our evaluation codes, synthesized dataset, and pre-trained weights will be publicly released, establishing a principled and scalable pathway toward more robust multimodal retrieval systems.

CHAPTER 4

DEMYSTIFYING SPATIOTEMPORAL SCALING LAWS OF MULTIMODAL VIDEO EMBEDDINGS

Chapter Abstract. As text-video retrieval becomes essential for large-scale video applications, models are expected to handle a diverse range of queries, from broad thematic summaries to descriptions of fleeting moments. However, since long videos are typically processed via sparse sampling, it remains unclear how sampling density affects retrieval across varying granularities. To address this, we introduce MG-LVR, a benchmark of multi-granular queries over a long-video corpus, constructed via a sparse-to-dense, semantics-aware pipeline. Evaluation across 23 multimodal embedding models reveals a significant granularity gap: fine-grained retrieval lags coarse-grained performance by over 75% under sparse sampling. Our analysis of spatiotemporal scaling shows that while spatial resolution is most beneficial at low frame counts, the optimal token allocation shifts systematically with query granularity. We further observe a training-time asymmetry where temporal representations are heavily anchored to specific frame counts. Leveraging these insights, we propose a progressive dense-to-sparse self-distillation framework that improves fine-grained query performance by up to 20%. This work provides a granularity-aware perspective on spatiotemporal scaling, offering principled designs for more robust video embedding training and inference.

4.1 Introduction

Text-video retrieval has been studied primarily in settings where a query summarizes the entirety of a short clip [23]. However, real-world videos are increasingly long and informationally dense, spanning tens of minutes to hours [19, 20]. Within such videos, a user query may describe anything from the overall theme to a brief, specific moment, making retrieval

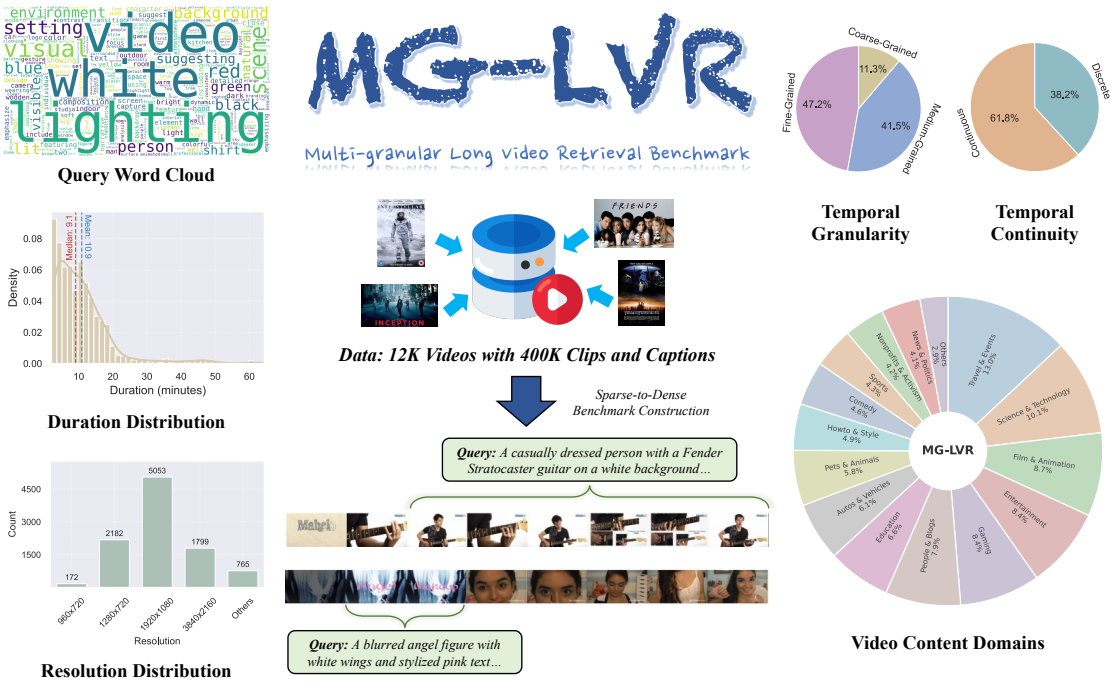


Figure 4.1: Overview of Multi-granular Long Video Retrieval (MG-LVR) benchmark.

inherently multi-granular [21, 22]. Since encoding every frame at full resolution is computationally prohibitive [16], models must rely on spatiotemporal sampling to produce video embeddings, which is a process proven critical for retrieval quality [17]. The sampling strategy thus determines the available visual evidence. Too few frames or too low a resolution can cause the target moment to be missed entirely, constituting a structural information loss. Therefore, how sampling density interacts with query granularity is a central question for long-video retrieval. Nevertheless, while previous works have demonstrated the sensitivity of performance to sampling [17, 24], this relationship remains underexplored because no existing benchmark provides the multi-granular and controllable testbed required to isolate these effects.

Current benchmarks are insufficient for this purpose. Short-clip datasets such as MSR-VTT [6] and DiDeMo [18] are too brief for sampling density to have a meaningful impact. Recent long-video retrieval benchmarks [11] extend the temporal scope but evaluate at a single query granularity, making it impossible to determine whether sampling configurations

should vary with query specificity. No existing setting jointly provides long-form videos and multi-granular query categorization within a unified evaluation framework. Such a combination is essential for exploring spatiotemporal scalability across diverse semantic scales.

To address this gap, we introduce the Multi-Granular Long Video Retrieval (MG-LVR) benchmark shown in Figure 4.1. MG-LVR pairs long-form videos with queries categorized by temporal granularity, defined as the fraction of the video duration that a query covers. Importantly, the benchmark allows the relevant evidence for a query to be grounded either in a single clip or across multiple clips that may occur at different timestamps, reflecting realistic retrieval scenarios. Evaluation across 23 models reveals a pronounced granularity gap: fine-grained retrieval lags coarse-grained performance by over 75% under sparse sampling and recovers only marginally as frame count increases. This provides a clear signal for studying how sampling configurations affect retrieval performance across different semantic granularities.

Using MG-LVR, we conduct a comprehensive spatiotemporal scaling analysis across both Contrastive Language-Image Pretraining (CLIP)-based [38] and Multimodal Large Language Model (MLLM)-based families [4]. At test time, increasing spatial resolution helps most when frames are few, while its benefit saturates at higher frame counts. Under a fixed token budget, the optimal spatial-temporal allocation shifts with query granularity. Coarse queries favor fewer frames at higher resolution, while fine-grained queries favor more frames at lower resolution. At training time, the interaction between spatial and temporal density is asymmetric. Spatial representations are resolution-robust, maintaining accuracy across varying input resolutions. Temporal representations exhibit frame-count anchoring, where performance peaks when the inference frame count matches the training configuration and degrades significantly when they diverge, particularly for fine-grained queries.

Table 4.1: Comparison of text-video retrieval benchmarks. **MG-LVR** is the first benchmark supporting multi-granular queries for retrieving long-form videos. **Gran.**: Temporal granularity of queries, including fine (FN), medium (MD), and coarse (CR) -grained; **Cont.**: Temporal continuity of queries, including discrete (DC) and continuous (CT).

Dataset	**# Query**	**# Corpus**	**Dur (s)**	**# Word**	**Gran.**	**Cont.**
MSRVTT [6]	1,000	1,000	15.0	9.4	CR	CT
DiDeMo [18]	1,004	1,004	53.9	29.1	CR	CT
TVR [50]	10,895	2,179	76.2	13.4	FN, MD	CT
ActivityNet Captions [51]	18,000	4,917	117.6	14.8	FN, MD	CT
LoVR [11]	100	467	1560.3	17364.5	CR	CT
VDC [15]	1,000	1,027	30.1	508.0	CR	CT
MG-LVR (Ours)	9,971	4,844	705.9	44.7	All	Both

This analysis highlights a key insight that the knowledge required for effective low-density inference is already latent within high-density models, but it does not automatically transfer when the sampling configuration changes. To address this, we propose Video Embedding Waterfall (VEterfall), a progressive dense-to-sparse self-distillation framework. Inspired by the waterfall effect, our method cascades high-fidelity semantic knowledge from densely sampled embeddings down to sparser representations. In this way, VEterfall achieves up to a 20% relative recall gain on fine-grained queries while operating in a $4\times$ to $16\times$ more efficient inference setting. Our experiments confirm that the scaling behaviors identified by MG-LVR can be directly and effectively exploited in practice.

In summary, our contributions are: (1) MG-LVR, the first benchmark for multi-granular long-video retrieval that enables scalable research into spatiotemporal sampling; (2) a multi-dimensional scaling analysis identifying key principles such as frame-count anchoring and granularity-dependent token allocation; and (3) VEterfall, a self-distillation method that enhances video embeddings under sparse sampling by unlocking multi-granular knowledge already latent within the models.

4.2 Related Works

4.2.1 Video Retrieval Benchmarks

Early text-video retrieval benchmarks such as MSR-VTT [6] and DiDeMo [18] consist of short clips in which sparse frame sampling suffices to capture the full content, rendering sampling strategy largely inconsequential. Subsequent benchmarks have extended the scope in complementary directions: LoVR [11] targets retrieval over hour-long videos, while TVR [50] and ActivityNet Captions [51] provide fine-grained temporal annotations for partially relevant video retrieval [52]. Despite this progress, each benchmark addresses only one dimension of the problem, covering either video duration or query granularity, but not both. MG-LVR addresses this gap by pairing a long-form corpus with temporally categorized queries, enabling the first controlled study of how spatiotemporal sampling interacts with retrieval granularity.

4.2.2 Video Embedding Models

Video-language embedding models have evolved from frame-level temporal aggregation [3] to dedicated spatiotemporal encoders, dividing into two families. CLIP-based models extend contrastive image-text pretraining to video, with representative examples including InternVideo2 [13] and X-CLIP [38]. MLLM-based models repurpose large multimodal language models for embedding extraction, as exemplified by GVE [17], Qwen3-VL-Embedding [46], and other modality-unified approaches [4]. Scaling-law studies in this domain have primarily examined model size and training data volume; a recent complementary effort [24] establishes power-law scaling with respect to frame count and caption density on short clips. The present work differs by evaluating end-to-end multimodal embedding models on long-form video, decomposing sampling into orthogonal spatial and temporal axes, and introducing query granularity as a first-class variable.

Building on the resulting scaling principles, we further propose VEterfall, a progressive dense-to-sparse self-distillation framework distinguished from conventional knowledge distillation by its cascaded, density-descending regime.

4.3 Rethinking Text-Video Retrieval Evaluation

Text-video retrieval has historically been formulated as a global-matching task, where each query describes the entirety of its paired video. This convention reflects the structural limitations of existing benchmarks: queries are either brief summaries of short clips or document-level aggregations of dense captions, making the query and video co-extensive by construction and effectively sidestepping the challenge of temporal scale. In the era of long, multi-scene video, this paradigm is insufficient. User intent may range from identifying a moment of a few seconds within an hour-long lecture to capturing the overarching theme of an entire documentary, imposing qualitatively different demands on spatiotemporal sampling and representation density.

As shown in Figure 4.1, the **Multi-Granular Long Video Retrieval (MG-LVR)** benchmark is designed to bridge this gap by formalizing *temporal granularity* as an explicit evaluation axis. Unlike traditional datasets [50, 51] where granularity emerges as a byproduct of annotator bias and video readability (e.g., length, dynamics, complexity), MG-LVR employs an automated pipeline that explicitly controls query granularity by analyzing semantic relationships among clips within each video, enabling a systematic study of how models navigate the spectrum from localized events to global narratives. The resulting benchmark contains $9,971$ queries paired with a corpus of $4,844$ videos averaging 705.9 seconds in duration. As summarized in Table 4.1, MG-LVR is the only benchmark that jointly covers all three granularity levels while supporting both single-clip and multi-clip evidence structures. The dataset and code of MG-LVR will be publicly released upon acceptance.

4.3.1 Formulations

Definition 4.3.1 (Text-Video Retrieval) *Given a text query q and a video corpus $\mathcal{V}$, the task is to identify the target video $v^* \in \mathcal{V}$ whose content is described by q. The query may refer to any portion of the video, from a brief localized event to the overall theme, with the fraction of relevant content serving as a variable difficulty factor.*

Definition 4.3.2 (Temporal Granularity) *Temporal granularity G is defined as the ratio of relevant content duration to total video length:*

$$G = \frac{|\mathcal{S}(v^*,\, q)|}{d_{\text{video}}} \in (0,\, 1]. \tag{4.1}$$

As $G \to 0$, retrieval approaches a needle-in-a-video problem in which the target moment occupies a vanishingly small fraction of the video. For systematic evaluation, we partition the granularity spectrum into three categories: ***Fine-grained (FN)****,* ***Medium-grained (MD)****, and* ***Coarse-grained (CR)****.*

In this work, granular thresholds are set to $G \in [0.001, 0.01]$ for FN and $G \in [0.1, 0.3]$ for MD, ensuring high inter-class distinctiveness and enabling clear observation of how model performance scales with task difficulty.

For the temporal distribution of relevant content, we further define temporal continuity C to characterize the structural complexity of each retrieval target:

$$C = \frac{\sum_{i=1}^{k}(e_i - s_i)}{e_k - s_1} \in (0,\, 1], \tag{4.2}$$

where $[s_i, e_i]$ are the k disjoint intervals composing $\mathcal{S}$. A value of $C = 1$ indicates a single contiguous event, while $C < 1$ represents fragmented content distributed across multiple disjoint segments, such as a recurring theme.

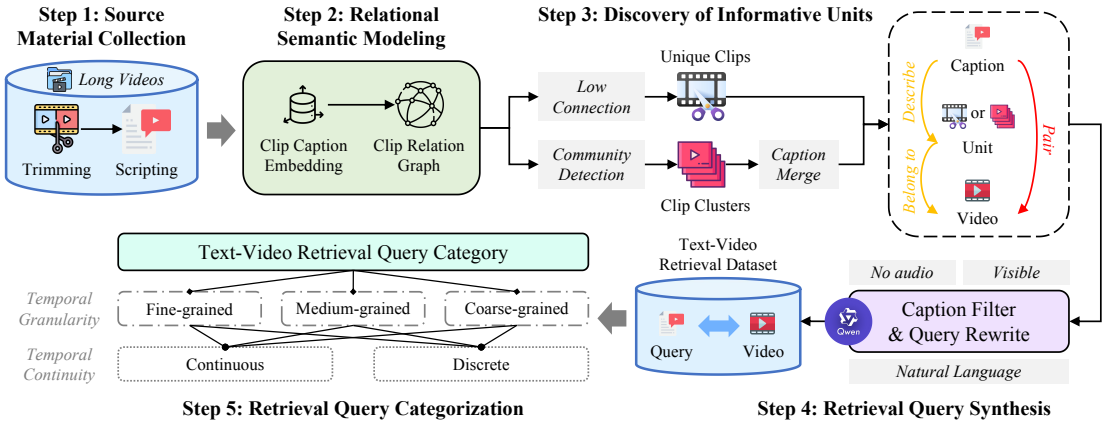


Figure 4.2: The MG-LVR construction pipeline. We model long videos as *Clip Relation Graphs* to transform unstructured content into a quantifiable semantic topology. This structural foundation allows for the systematic discovery of retrieval units across varying temporal granularities and continuity levels, which are then synthesized by an LLM into natural language queries.

4.3.2 MG-LVR: Sparse-to-Dense Benchmark Construction

We construct MG-LVR through a semantics-aware pipeline that formalizes video retrieval as a sampling task over structured semantic relationships among clips. The core insight is to represent each video as a similarity graph, enabling the identification of retrieval units that vary in temporal granularity and evidence composition. This approach ensures that MG-LVR covers scenarios where relevant evidence is grounded either in a single clip or across multiple clips that may occur at different timestamps.

Step 1: Material Curation. We build upon the Vript dataset [69], selecting long videos ($>$ 120s) with dense, high-quality GPT-4V clip-level captions. These captions serve as semantic proxies, allowing the pipeline to model video content without the prohibitive cost of pixel-level processing during construction.

Step 2: Semantic Topology Modeling. To provide a structural basis for unit extraction, each video is represented as a *Clip Relation Graph*, where nodes correspond to clips, and edges correspond to the cosine similarity values between two clip caption embeddings

produced by Qwen3-Embedding-8B [70]. This graph captures semantic relationships among clips: clusters of highly connected nodes often correspond to dominant themes, while weaker connections may indicate localized or transient events.

Step 3: Discovery of Informative Units. Retrieval units are extracted from the clip relation graph by applying structural constraints that correspond to our granularity and evidence composition settings. This graph-based discovery allows the benchmark to include both single-clip evidence and semantically related clips that appear at different timestamps. *Single-Clip Units:* Individual nodes with nearly zero external connections are selected as standalone retrieval units, yielding $32,984$ units. *Multi-Clip Units:* Identified via Louvain community detection to capture thematically consistent clips that may be dispersed across the timeline, yielding $7,764$ units.

Step 4: Retrieval Query Synthesis. An LLM (Qwen3-32B [71]) synthesizes the metadata of each unit into a natural language query. For multi-clip units, the model is prompted to generate a holistic description covering the relationships among multiple relevant clips. A subsequent refinement pass filters out ambiguous descriptions and removes references to audio content to ensure a focus on visual information.

Step 5: Retrieval Query Categorization. Each query-video pair is assigned to a granularity level (FN, MD, or CR) based on the computed value of G. The final benchmark retains $9,971$ queries in total, comprising $4,706$, $4,142$, and $1,123$ queries for FN, MD, and CR, respectively.

Human Validation. To ensure retrieval uniqueness, a cross-modal collision check is applied: queries exhibiting high semantic similarity to non-target clips within the same video are pruned. Human validation confirms a 96% accuracy rate over 50 query-video pairs, with

Table 4.2: Multi-granularity retrieval performance of all baselines at 8 uniformly sampled frames per video. **Bold** and underline indicate the best and second-best results within each model group, respectively.

Model	Overall		FN		MD		CR	
	R@1	MRR	R@1	MRR	R@1	MRR	R@1	MRR
InternVideo2-1B	0.2398	0.3216	0.0803	0.1383	0.3390	0.4435	0.5423	0.6398
ViCLIP	0.2659	0.3491	0.1054	0.1714	0.3790	0.4772	0.5209	0.6207
CLIP4Clip	0.2696	0.3561	0.1067	0.1703	0.3660	0.4757	0.5966	0.6938
InternVideo2-6B	0.2890	0.3757	0.1069	0.1756	0.4029	0.5093	0.6322	0.7215
VideoCLIP-XL	0.3914	0.4794	0.1687	0.2488	0.5449	0.6466	**0.7587**	**0.8293**
LanguageBind	**0.4194**	**0.5046**	**0.1997**	**0.2811**	**0.5787**	**0.6728**	0.7524	0.8204
GME-2B	0.2793	0.3642	0.1128	0.1783	0.3754	0.4816	0.6224	0.7101
Unite-2B	0.2875	0.3728	0.1169	0.1834	0.3889	0.4936	0.6287	0.7213
Ops-MM-2B	0.3085	0.3893	0.1258	0.1918	0.4211	0.5180	0.6589	0.7416
VLM2Vec-V2	0.3089	0.3870	0.1243	0.1833	0.4162	0.5147	0.6866	0.7695
GVE-3B	0.3747	0.4593	0.1583	0.2358	0.5128	0.6100	**0.7720**	**0.8397**
Qwen3-VL-Emb-2B	**0.3914**	**0.4751**	**0.1723**	**0.2481**	**0.5386**	**0.6348**	0.7667	0.8375
U-MARVEL	0.1169	0.1919	0.0368	0.0784	0.1526	0.2501	0.3215	0.4528
LamRA-Ret	0.1885	0.2671	0.0858	0.1414	0.2566	0.3516	0.3678	0.4822
GME-7B	0.3161	0.4018	0.1309	0.2004	0.4278	0.5326	0.6803	0.7630
BGE-VL	0.3201	0.4000	0.1362	0.2018	0.4425	0.5376	0.6394	0.7228
B3-7B	0.3475	0.4282	0.1519	0.2211	0.4657	0.5627	0.7311	0.7999
Ops-MM-7B	0.3546	0.4357	0.1538	0.2258	0.4778	0.5733	0.7418	0.8076
RzenEmbed	0.3707	0.4541	0.1498	0.2246	0.5118	0.6101	0.7756	0.8402
Unite-7B	0.3733	0.4591	0.1664	0.2453	0.5007	0.6011	0.7703	0.8312
UniME-V2	0.3881	0.4721	0.1575	0.2338	0.5447	0.6415	0.7774	0.8458
GVE-7B	0.4121	0.4960	0.1813	0.2634	0.5652	0.6590	0.8148	0.8693
Qwen3-VL-Emb-8B	**0.4492**	**0.5269**	**0.2102**	**0.2861**	**0.6106**	**0.6993**	**0.8557**	**0.9002**

failure cases detailed in the supplementary material.

4.4 Spatiotemporal Scaling Laws: Experiments, Analysis, and Discoveries

Leveraging MG-LVR as a controlled testbed, we systematically analyze how spatiotemporal sampling density affects retrieval performance across query granularity levels. The analysis proceeds in three phases: (i) baseline comparison under default configurations; (ii) test-time scaling, in which spatial and temporal densities are varied during inference; and (iii) training-time scaling, in which the density used during model training is adjusted to examine

Table 4.3: Multi-granularity retrieval performance of MLLM-based baselines at 128 uniformly sampled frames per video. **Bold** and underline indicate the best and second-best results within each model group, respectively.

Model	Overall		FN		MD		CR	
	R@1	MRR	R@1	MRR	R@1	MRR	R@1	MRR
GME-2B	0.3576	0.4470	0.1575	0.2378	0.4947	0.5974	0.6910	0.7689
Unite-2B	0.3802	0.4673	0.1698	0.2494	0.5321	0.6305	0.7017	0.7783
Ops-MM-2B	0.4098	0.4956	0.1872	0.2729	0.5751	0.6651	0.7329	0.8040
VLM2Vec-V2	0.4369	0.5190	0.2129	0.2926	0.6045	0.6936	0.7569	0.8236
Qwen3-VL-Emb-2B	0.4901	0.5710	0.2318	0.3227	**0.6849**	**0.7640**	**0.8540**	**0.9000**
GVE-3B	**0.4914**	**0.5732**	**0.2433**	**0.3314**	0.6801	0.7624	0.8353	0.8881
GME-7B	0.4171	0.5057	0.1785	0.2651	0.5942	0.6909	0.7640	0.8307
UniME-V2	0.4461	0.5290	0.2004	0.2841	0.6260	0.7146	0.8121	0.8705
Ops-MM-7B	0.4708	0.5519	0.2316	0.3193	0.6497	0.7320	0.8130	0.8628
Unite-7B	0.4736	0.5584	0.2340	0.3244	0.6497	0.7379	0.8281	0.8773
GVE-7B	0.5075	0.5869	0.2663	0.3567	0.6900	0.7662	0.8451	0.8900
RzenEmbed	0.5112	0.5889	0.2401	0.3290	0.7228	0.7972	0.8664	0.9095
Qwen3-VL-Emb-8B	**0.5488**	**0.6220**	**0.2871**	**0.3786**	**0.7501**	**0.8137**	**0.9029**	**0.9355**

its effect on the quality of the learned embeddings.

4.4.1 Experimental Setup

Model Selection. Twenty-three models from two paradigms are evaluated: 6 CLIP-based models and 17 MLLM-based models.

CLIP-based models: CLIP4Clip [3] (87M); ViCLIP [65] (0.4B); VideoCLIP-XL [39] (0.4B); LanguageBind-Video-Huge-V1.5 [40] (1.2B); InternVideo2-Stage2-1B [13] (1.4B); InternVideo2-Stage2-6B [13] (6.4B).

MLLM-based models: gme-Qwen2-VL-2B-Instruct [12] (GME-2B, 2.2B); Unite-Base-Qwen2-VL-2B [4] (Unite-2B, 2.2B); VLM2Vec-V2.0 [42] (VLM2Vec-V2, 2.2B); Ops-MM-embedding-v1-2B [72] (Ops-MM-2B, 2.2B); GVE-3B [17] (GVE-3B, 3.8B); Qwen3-VL-Embedding-2B [46] (Qwen3-VL-Emb-2B, 2.1B); BGE-VL-v1.5-mmeb [43] (BGE-VL, 7.6B); UniME-V2-LLaVA-OneVision-8B [73] (UniME-V2, 8.0B); B3-Qwen2-7B [45] (B3-7B, 8.3B); gme-Qwen2-VL-7B-Instruct [12] (GME-7B, 8.3B); Unite-Base-Qwen2-VL-7B [4] (Unite-7B, 8.3B); Ops-MM-embedding-v1-7B [72] (Ops-MM-7B,

8.3B); U-MARVEL-Qwen2VL-7B-Instruct [74] (U-MARVEL, 8.3B); LamRA-Ret-Qwen2.5VL-7B [75] (LamRA-Ret, 8.3B); GVE-7B [17] (GVE-7B, 8.3B); RzenEmbed-7B [76] (RzenEmbed, 8.3B); Qwen3-VL-Embedding-8B [46] (Qwen3-VL-Emb-8B, 8.1B).

Evaluation Protocol. Retrieval performance is measured using Recall@1 (R@1) and Mean Reciprocal Rank (MRR) across three query granularity levels: Fine-grained (FN), Medium-grained (MD), and Coarse-grained (CR). The total visual token budget per video is defined as $N_{\text{total}} = F \times T$, where F denotes the number of uniformly sampled frames and T denotes the number of spatial tokens per frame as determined by the visual encoder. In the analyses that follow, F and T are varied independently while all other experimental factors are held constant to isolate the effect of sampling density.

4.4.2 Baseline Comparison

All twenty-three models are evaluated under two configurations (8 and 128 frames; results at 32 frames are provided in the supplementary material). As shown in Table 4.2 and Table 4.3, performance degrades consistently from CR to FN across all architectures, regardless of model family, parameter scale, or training data composition. For Qwen3-VL-Emb-8B, R@1 drops from 0.8557 on CR to 0.2102 on FN at 8 frames, a relative decline of over 75%. Even at 128 frames, FN R@1 improves only modestly to 0.2871, a gain of less than 8 percentage points despite a $16\times$ increase in temporal sampling density. This pronounced granularity gradient indicates that simply increasing the number of frames yields strongly diminishing returns on fine-grained retrieval.

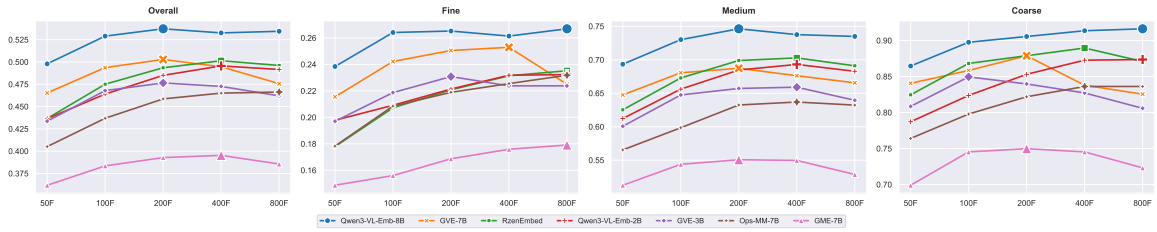


Figure 4.3: Test-time spatial scaling results at 32 frames per video. Sensitivity results under varying temporal densities are provided in the supplementary material.

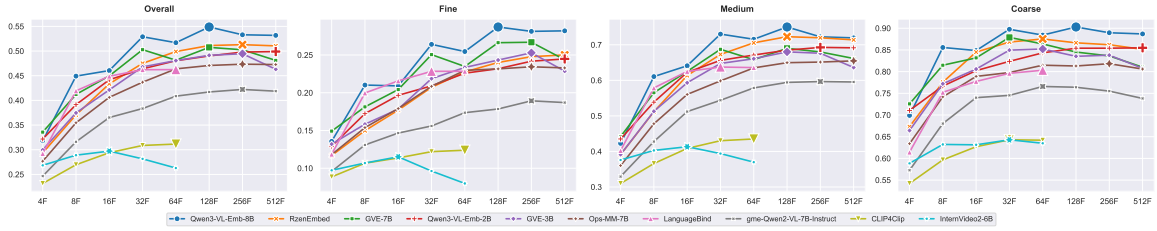


Figure 4.4: Test-time temporal scaling across model families. MLLM-based models exhibit continuous improvement with increasing frame count, whereas CLIP-based models plateau and degrade at high frame counts.

4.4.3 Test-Time Scaling

We examine how varying spatiotemporal sampling density at inference time affects retrieval performance, isolating spatial and temporal effects by fixing one dimension while varying the other.

Spatial Scaling. For MLLM-based models, the benefit of increasing spatial tokens per frame depends critically on the available temporal context (Figure 4.3). When the number of frames is low ($F \leq 8$), higher spatial resolution yields consistent improvements. As F increases beyond 32, the marginal gain from additional spatial tokens diminishes and occasionally reverses, particularly for CR queries where holistic scene understanding is more important than fine spatial detail. This suggests that spatial and temporal information are complementary rather than independently additive.

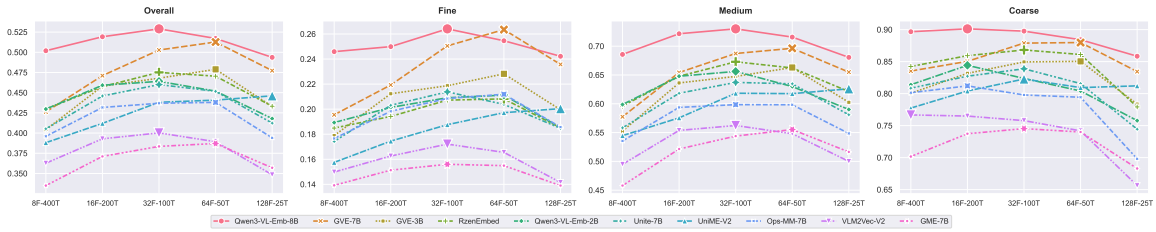


Figure 4.5: Retrieval performance under a fixed total token budget. The optimal allocation between spatial and temporal tokens shifts systematically with query granularity.

Temporal Scaling. Varying F from 4 to 512 frames reveals a divergence in scaling behavior across model families (Figure 4.4). MLLM-based models exhibit continuous improvement up to $F = 128$, with the largest absolute gains on FN queries. CLIP-based models, by contrast, plateau around $F = 16$ and exhibit performance degradation on FN and MD queries at higher frame counts. This difference is consistent with the underlying architectural design: MLLM-based models process frames jointly through a transformer with positional encodings, whereas CLIP-based models aggregate frame-level features via mean pooling, which dilutes discriminative signals as F increases.

Spatiotemporal Trade-off. When N_{total} is held constant, the optimal allocation between F and T depends on query granularity (Figure 4.5). CR queries favor high-resolution, low-frame-count configurations, while FN queries favor more frames at lower spatial resolution. Reducing spatial tokens primarily discards fine-grained texture while preserving semantic content, whereas reducing the number of frames risks eliminating visual evidence of brief events entirely.

4.4.4 Training-Time Scaling

Following the protocol of an existing work [17], we post-train Qwen2.5-VL-Instruct-3B [67] with InfoNCE loss [68] on 500K instances to examine how the training density $(F_{\text{train}}, T_{\text{train}})$ shapes the learned embedding space.

Table 4.4: Training-time spatial scaling results at a fixed temporal density of 4 frames. Models trained at varying spatial resolutions ($T_{\text{train}} \in \{50, 100, 200, 400\}$) are evaluated across all spatial test configurations. Bold indicates the best result within each group.

Training		Testing		Overall		FN		MD		CR	
Temporal	Spatial	Temporal	Spatial	R@1	MRR	R@1	MRR	R@1	MRR	R@1	MRR
4F	50T	4F	50T	0.1941	0.2679	0.0731	0.1226	0.2591	0.3509	0.4613	0.5710
	100T			0.2019	0.2770	0.0797	0.1298	0.2677	0.3617	0.4711	0.5811
	200T			0.2060	0.2820	0.0837	0.1346	0.2714	0.3671	0.4773	0.5865
	400T			**0.2096**	**0.2851**	**0.0865**	**0.1378**	**0.2740**	**0.3687**	**0.4880**	**0.5939**
4F	50T	4F	100T	0.2243	0.3011	0.0865	0.1390	0.2996	0.3968	0.5236	0.6277
	100T			0.2212	0.2979	0.0837	0.1375	0.2945	0.3901	0.5272	0.6299
	200T			0.2384	0.3180	0.0922	0.1494	0.3182	0.4173	0.5565	0.6583
	400T			**0.2474**	**0.3262**	**0.0975**	**0.1558**	**0.3300**	**0.4267**	**0.5708**	**0.6696**
4F	50T	4F	200T	0.2406	0.3168	0.0890	0.1420	0.3199	0.4178	0.5833	0.6764
	100T			0.2425	0.3202	0.0922	0.1473	0.3197	0.4187	0.5877	0.6820
	200T			0.2454	0.3230	0.0948	0.1500	0.3238	0.4220	0.5877	0.6834
	400T			**0.2581**	**0.3385**	**0.1031**	**0.1628**	**0.3440**	**0.4425**	**0.5913**	**0.6910**
4F	50T	4F	400T	0.2503	0.3263	0.0926	0.1471	0.3353	0.4318	**0.5975**	0.6883
	100T			0.2507	0.3285	0.0914	0.1501	0.3378	0.4332	**0.5975**	0.6898
	200T			0.2561	0.3342	**0.0984**	**0.1566**	0.3431	0.4392	0.5966	**0.6909**
	400T			**0.2575**	**0.3359**	0.0963	0.1563	**0.3496**	**0.4442**	0.5939	0.6889

Spatial Scaling: Resolution Robustness. Increasing T_{train} yields consistent performance gains across most granularity levels and inference resolutions (Table 4.4). Models trained at $T_{\text{train}} = 400$ achieve the best Overall R@1 in all four test-time spatial groups. Notably, even when evaluated at $T_{\text{test}} = 50$, the model trained at $T_{\text{train}} = 400$ substantially outperforms the model trained directly at $T_{\text{train}} = 50$ ($+0.0155$ Overall R@1). We term this property *resolution robustness*: representations learned at high spatial density generalize effectively to low-density inference. However, this trend is not strictly monotonic at all granularity levels: at $T_{\text{test}} = 400$, increasing T_{train} from 200 to 400 yields only marginal overall gains ($+0.0014$ Overall R@1) while slightly degrading Fine R@1 (-0.0021) and Coarse R@1 (-0.0027), suggesting diminishing returns at extreme spatial density.

Temporal Scaling: Frame-Count Anchoring. Unlike spatial representations, temporal representations are tightly coupled to the frame count seen during training, a phenomenon we term *frame-count anchoring*. A mismatch between F_{train} and F_{test} consistently degrades

Table 4.5: Training-time temporal scaling results at a fixed spatial density of 50 tokens per frame. Models trained at varying frame counts ($F_{\text{train}} \in \{4, 8, 16, 32\}$) are evaluated across six temporal test configurations. Bold indicates the best result within each group.

Training		Testing		Overall		FN		MD		CR	
Temporal	Spatial	Temporal	Spatial	R@1	MRR	R@1	MRR	R@1	MRR	R@1	MRR
4F	50T	4F	50T	0.2076	0.2837	0.0841	0.1352	**0.2706**	**0.3666**	0.4924	0.6002
8F				0.2080	0.2842	0.0858	0.1374	0.2694	0.3654	**0.4933**	**0.6004**
16F				**0.2082**	**0.2854**	**0.0882**	**0.1396**	0.2677	0.3659	0.4915	0.5987
32F				0.1930	0.2687	0.0737	0.1261	0.2528	0.3466	0.4720	0.5792
4F	50T	8F	50T	0.2639	0.3515	0.1054	0.1694	0.3566	0.4669	0.5859	0.6892
8F				**0.2741**	**0.3605**	**0.1143**	**0.1788**	**0.3667**	**0.4764**	**0.6020**	**0.6946**
16F				0.2486	0.3295	0.0992	0.1591	0.3281	0.4291	0.5815	0.6763
32F				0.2390	0.3168	0.0982	0.1531	0.3117	0.4095	0.5610	0.6614
4F	50T	16F	50T	0.3165	0.4086	0.1207	0.1951	0.4464	0.5590	0.6581	0.7484
8F				0.3071	0.3941	0.1147	0.1839	0.4288	0.5363	0.6643	0.7501
16F				**0.3254**	**0.4136**	**0.1235**	**0.1972**	**0.4587**	**0.5648**	**0.6803**	**0.7628**
32F				0.3090	0.3987	0.1196	0.1903	0.4254	0.5385	0.6732	0.7560
4F	50T	32F	50T	0.3310	0.4196	0.1373	0.2088	0.4611	0.5678	0.6625	0.7559
8F				0.3556	0.4433	0.1432	0.2192	0.5017	0.6057	0.7070	0.7831
16F				0.3574	0.4466	**0.1466**	**0.2255**	0.5034	0.6068	0.7026	0.7820
32F				**0.3596**	**0.4476**	0.1456	0.2224	**0.5075**	**0.6118**	**0.7115**	**0.7857**
4F	50T	64F	50T	0.3829	0.4717	0.1655	0.2440	0.5328	0.6392	**0.7409**	**0.8110**
8F				0.3772	0.4640	0.1596	0.2375	0.5280	0.6294	0.7329	0.8031
16F				0.3877	0.4746	**0.1698**	**0.2488**	0.5420	0.6421	0.7320	0.8026
32F				**0.3892**	**0.4769**	0.1668	0.2472	**0.5468**	**0.6472**	0.7400	**0.8110**
4F	50T	128F	50T	0.4021	0.4900	0.1772	0.2604	0.5633	0.6626	0.7498	0.8174
8F				0.3941	0.4799	0.1698	0.2511	0.5538	0.6500	0.7453	0.8116
16F				0.4058	0.4939	0.1806	0.2651	**0.5681**	**0.6661**	0.7507	0.8174
32F				**0.4078**	**0.4949**	**0.1859**	**0.2695**	0.5666	0.6629	**0.7516**	**0.8195**

performance in both directions, but the degradation is asymmetric. When $F_{\text{train}} \gg F_{\text{test}}$, the performance drop is more severe: a model trained at $F_{\text{train}} = 32$ and evaluated at $F_{\text{test}} = 4$ drops 0.0104 in Fine R@1 relative to the matched $F_{\text{train}} = 4$ baseline (Group 1 in Table 4.5). In the inverse direction, a model trained at $F_{\text{train}} = 4$ and evaluated at $F_{\text{test}} = 32$ drops 0.0083 in Fine R@1 relative to the matched $F_{\text{train}} = 32$ baseline (Group 4 in Table 4.5). Although these gaps are measured against different absolute performance levels and should be interpreted accordingly, the consistent pattern across groups suggests that low-frame-count training better preserves fine-grained discriminability under temporal mismatch than high-frame-count training does.

4.5 VEterfall: Channeling Scaling Potential for Self-Enhancement of Video Embedding Models

Our analysis reveals a key asymmetry: spatial embeddings tolerate token compression well, whereas temporal ones are governed by frame-count anchoring, i.e., models evaluated with high frame-density inputs degrade when the frame count is reduced at inference time. This suggests that the knowledge required for low-density inference is already latent within representations produced under high-density input sampling, but is not automatically transferred when the sampling configuration changes. We therefore propose **Video Embedding Waterfall (VEterfall)**, a progressive dense-to-sparse self-distillation framework that explicitly propagates this knowledge along the sampling-density axis, from high- to low-density configurations stage by stage, analogous to water descending through successive levels of a waterfall as shown in Figure 4.6.

4.5.1 Method

Single Dense-to-Sparse Distillation Step. The basic unit of VEterfall is a single distillation step [58] from a denser to a sparser input sampling configuration, where density refers jointly to the number of frames sampled from the video and the number of spatial tokens retained per frame at inference time. A frozen teacher processes inputs at higher sampling density, while a student—sharing the same architecture and initialized from the same pre-trained checkpoint $\mathcal{V}^{(0)}$ (here GVE-3B [17])—is trained at a strictly lower density. The student is optimized to align its video embedding with that of the teacher via MSE loss, with an additional InfoNCE [68] loss term (λ=0.5) to maintain video-text alignment by contrastive learning throughout density reduction.

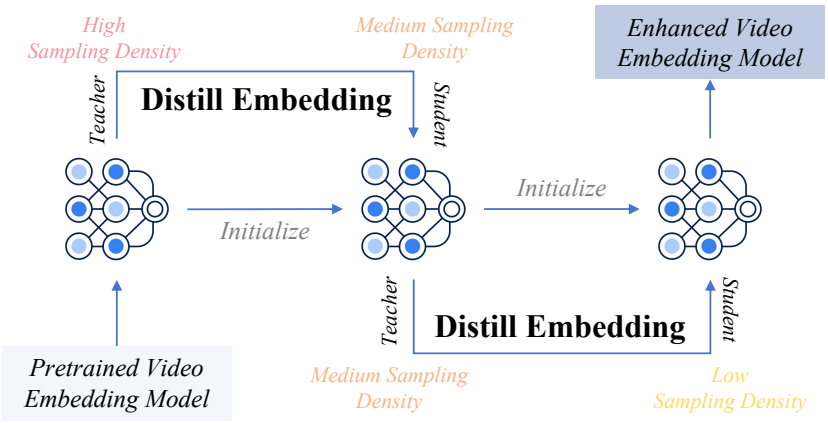


Figure 4.6: VEterfall progressively distills video embeddings across progressively sparser input sampling configurations using the pretrained model itself as the teacher at each stage.

K-Round Cascade. To support large overall density reductions while keeping each individual step within a reliable transfer range, VEterfall chains K distillation steps into a cascade, where each round takes the previous checkpoint as its frozen teacher: $\mathcal{V}^{(k)} \xleftarrow{\text{distill}} \mathcal{V}^{(k-1)},\ k = 1, \ldots, K$. Here $\mathcal{V}^{(0)}$ serves as the pre-trained video embedding model, and K controls the granularity of the reduction schedule.

Instantiation. We set K=2. Starting from $\mathcal{V}^{(0)}$ at (16F, 400T), the first round produces $\mathcal{V}^{(1)}$ at (8F, 200T), and the second produces $\mathcal{V}^{(2)}$ at (4F, 100T). Across two rounds, the input token budget is reduced by $16\times$ with no modification to the inference architecture. Each round uses 100K text-video pairs from InternVid [65], a substantially smaller data requirement than comparable models.

4.5.2 Experiments

Main Results. As shown in Table 4.6, each distillation round yields consistent R@1 gains, with improvements concentrated on fine- and medium-grained queries. This is consistent with our earlier finding that these granularities are most sensitive to frame sparsity. Coarse-grained performance remains stable across rounds, confirming that holistic scene under-

Table 4.6: **Iterative Distillation.** R@1 performance across successive rounds ($K = 1, 2$). VEterfall effectively recovers performance at lower densities, particularly for fine-grained queries.

Setting	Model	Overall	Fine	Medium	Coarse
16F-400T	$\mathcal{V}^{(0)}$	0.447	0.202	0.622	**0.829**
	$\mathcal{V}^{(1)}$	0.454	0.212	0.628	0.826
	$\mathcal{V}^{(2)}$	**0.456**	**0.215**	**0.630**	0.819
8F-200T	$\mathcal{V}^{(0)}$	0.369	0.148	0.508	**0.782**
	$\mathcal{V}^{(1)}$	0.398	0.182	0.540	0.780
	$\mathcal{V}^{(2)}$	**0.402**	**0.188**	**0.543**	0.777
4F-100T	$\mathcal{V}^{(0)}$	0.300	0.132	0.391	0.664
	$\mathcal{V}^{(1)}$	0.300	0.130	0.394	0.664
	$\mathcal{V}^{(2)}$	**0.325**	**0.158**	**0.423**	**0.667**

Table 4.7: **Frame-Efficiency.** Evaluated at fixed $T = 200$. VEterfall ($\mathcal{V}^{(1)}$) significantly improves performance as frame count F decreases.

Setting	Model	Overall	Fine	Medium	Coarse
16F-200T	$\mathcal{V}^{(0)}$	**0.458**	**0.212**	**0.636**	**0.832**
	$\mathcal{V}^{(1)}$	0.446	0.211	0.615	0.811
8F-200T	$\mathcal{V}^{(0)}$	0.369	0.148	0.508	**0.782**
	$\mathcal{V}^{(1)}$	**0.398**	**0.182**	**0.540**	0.780
4F-200T	$\mathcal{V}^{(0)}$	0.303	0.125	0.403	0.677
	$\mathcal{V}^{(1)}$	**0.319**	**0.138**	**0.420**	**0.706**

standing is already well-preserved at low density. At 4F-100T, $\mathcal{V}^{(2)}$ improves overall R@1 from 0.300 to 0.325, with a $\sim$20% relative gain on fine-grained queries. $\mathcal{V}^{(2)}$ also yields a marginal improvement at the original high-density setting, suggesting that the contrastive loss term provides mild regularization as a beneficial side effect.

Frame-Efficiency. We fix T=200 and vary the number of sampled frames F to isolate the effect of temporal sparsity and evaluate the effectiveness of $\mathcal{V}^{(1)}$. As summarized in Table 4.7, $\mathcal{V}^{(1)}$ consistently outperforms $\mathcal{V}^{(0)}$ at 8 frames and below, with gains growing as the frame budget tightens. This confirms that VEterfall is most effective in the low-frame regime relevant for efficient deployment.

Table 4.8: **Token-Efficiency.** Evaluated at fixed $F = 8$. Benefits are consistent but less pronounced than frame-count adjustments.

Setting	Model	Overall	Fine	Medium	Coarse
8F-400T	$\mathcal{V}^{(0)}$	0.404	0.179	0.553	**0.801**
	$\mathcal{V}^{(1)}$	**0.412**	**0.191**	**0.558**	0.800
8F-200T	$\mathcal{V}^{(0)}$	0.369	0.148	0.508	**0.782**
	$\mathcal{V}^{(1)}$	**0.398**	**0.182**	**0.540**	0.780
8F-100T	$\mathcal{V}^{(0)}$	0.375	0.158	0.513	**0.772**
	$\mathcal{V}^{(1)}$	**0.380**	**0.171**	**0.516**	0.758

Table 4.9: **Ablation on λ.** Impact of contrastive loss term weight on 8F-200T performance. $\lambda = 0.5$ provides the best alignment balance.

λ	Overall	Fine	Medium	Coarse
0	0.396	0.178	0.537	**0.782**
0.5	**0.398**	**0.182**	**0.540**	0.780

Token-Efficiency. As shown in Table 4.8, where we fix $F{=}8$ and vary T, $\mathcal{V}^{(1)}$ improves consistently across all spatial token budgets. However, the margins are smaller and more uniform compared to frame-efficiency gains, confirming that spatial token compression is a secondary factor relative to frame sparsity.

Ablation on Contrastive Learning. Table 4.9 illustrates the impact of the loss weight λ. Setting $\lambda{=}0$ already yields competitive performance, and adding the InfoNCE term provides marginal but consistent gains. These results confirm that the primary source of improvement is the embedding distillation objective rather than the contrastive learning term.

4.6 Conclusion

We introduced MG-LVR, a benchmark for the controlled study of spatiotemporal sampling across query granularity levels in long-video retrieval. Evaluating 23 models revealed a fundamental asymmetry between spatial and temporal scaling: spatial representations are resolution-robust, whereas temporal representations exhibit frame-count anchoring that is

most pronounced on fine-grained queries. Under a fixed token budget, the optimal allocation between spatial and temporal tokens depends on query granularity. Building on these findings, we proposed VEterfall, a progressive dense-to-sparse self-distillation framework in which the high-density representations of a model are used to improve its own low-density embeddings without requiring external supervision, achieving up to $\sim$20% relative R@1 gains on fine-grained queries under sparse sampling. An important direction for future work is to develop architectures and training strategies that generalize robustly across spatiotemporal sampling configurations.

CHAPTER 5

GPU-ACCELERATED MULTI-RELATIONAL PARALLEL GRAPH RETRIEVAL FOR WEB-SCALE RECOMMENDATIONS

Chapter Abstract. Web recommendations provide personalized items from massive catalogs for users, which rely heavily on retrieval stages to trade off the effectiveness and efficiency of selecting a small relevant set from billion-scale candidates in online digital platforms. As one of the largest Chinese search engine and news feed providers, Baidu resorts to Deep Neural Network (DNN) and graph-based Approximate Nearest Neighbor Search (ANNS) algorithms for accurate relevance estimation and efficient search for relevant items. However, current retrieval at Baidu fails in comprehensive user-item relational understanding due to dissected interaction modeling, and performs inefficiently in large-scale graph-based ANNS because of suboptimal traversal navigation and the GPU computational bottleneck under high concurrency. To this end, we propose a GPU-accelerated Multi-relational Parallel Graph Retrieval (GMP-GR) framework to achieve effective yet efficient retrieval in web-scale recommendations. First, we propose a multi-relational user-item relevance metric learning method that unifies diverse user behaviors through multi-objective optimization and employs a self-covariant loss to enhance pathfinding performance. Second, we develop a hierarchical parallel graph-based ANNS to boost graph retrieval throughput, which conducts breadth-depth-balanced searches on a large-scale item graph and cost-effectively handles irregular neural computation via adaptive aggregation on GPUs. In addition, we integrate system optimization strategies in the deployment of GMP-GR in Baidu. Extensive experiments demonstrate the superiority of GMP-GR in retrieval accuracy and efficiency. Deployed across more than twenty applications at Baidu, GMP-GR serves hundreds of millions of users with a throughput exceeding hundreds of millions of requests per second.

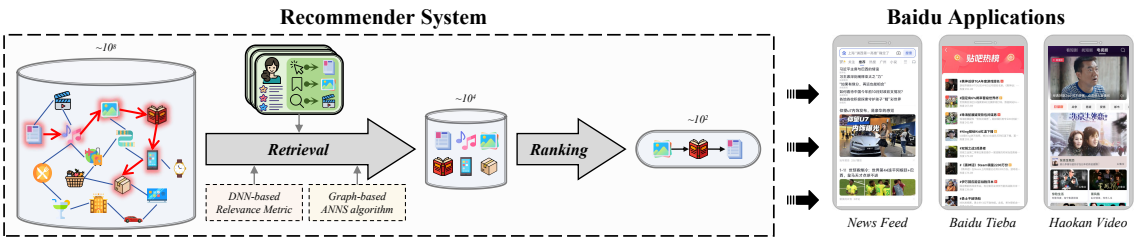


Figure 5.1: Applications of recommender systems at Baidu.

5.1 Introduction

Recently, recommender systems have become essential tools for helping users navigate various options and discover relevant products or content [25]. However, large-scale relevance evaluation based on user preference is computationally prohibitive in real-world recommendation applications [26]. For timely answering requirements, computing relevance values of billions of items is unacceptable [27]. To balance effectiveness and efficiency, retrieval-and-ranking recommendation frameworks [7, 77] have gained widespread adoption in modern industrial systems, such as YouTube [8], LinkedIn [78], and Pinterest [79]. The first stage (*i.e.* retrieval) focuses on retrieving a thousand-scale relevant subset from the entire billion-scale item catalog, which then serves as the candidates for further ordering in the second stage (*i.e.* ranking). In particular, retrieval plays an important role in real-time and satisfactory answers for large-scale queries. Effective retrieval can significantly simplify the search space for potential answers while maintaining personalization and serendipity, thus reducing the ranking difficulty in producing reliable final recommendations.

In this work, we focus on the retrieval stage in Baidu's web-scale recommendation scenarios, e.g., news feed, Tieba, and video clips. Recent works [29, 37] establish the retrieval framework in a training-and-search manner. Based on the complex relational dependencies modeling ability of Deep Neural Network (DNN) [80], they attempt to train a DNN-based metric for more accurate user-item relevance estimation compared with simple functions [35, 36]. Nevertheless, real-time retrieval is difficult due to complicated neural com-

putations on massive candidates. To solve the problem, researchers have been exploring the potential of Approximate Nearest Neighbor Search (ANNS) algorithms to mitigate extensive computations, which search for relatively relevant items for each query user by selectively visiting instead of the entire item base. Early works propose tree-based ANNS to organize data points in a tree-like structure by recursively splitting the dimensional space [81]. However, they underperform in web-scale recommendations that aim to retrieve divergent and high-dimensional information. Because they are sensitive to complex feature space and item relationships [33]. To improve retrieval effectiveness, the latest frameworks leverage graph-based ANNS algorithms that build similarity-based candidate graphs to efficiently expand the candidate set via graph traversal. For example, HNSW [28] hierarchically establishes a multi-layer indexing graph, grounding coarse-to-fine navigation towards the most satisfactory candidates.

However, we confront three challenges in developing an accurate and efficient retrieval framework for web-scale recommendations. *(1) Unified modeling of multi-relational user-item relevance.* Baidu recommendation applications consist of dozens of distinct and related behaviors (e.g., click, browse) that are typically modeled by a tailored metric for estimating relevance between users and items. However, the independent modeling of these user-item interactions fails to capture the semantic relations of diverse user behaviors, which are mutually beneficial between them [37]. Besides, training multiple behavior-specific metrics incurs exponential efforts in system updates. Thus, the first challenge lies in cost-effective multi-relational metric joint training for more accurate relevance estimation. *(2) Efficient search on large-scale graph item base.* Baidu devises Graph-based ANNS to traverse relevant candidate items from a graph item base. Practically, a trade-off problem arises that increasing the online retrieval throughput will restrict the search efforts, *i.e.* the graph traversal steps. However, existing methods make delivering satisfactory items in limited steps difficult, especially for billion-scale graphs in web-scale recommender systems. On the one

hand, neural relevance metrics often violate the triangle inequality, a property that ensures optimal pathfinding in graph traversal [29]. This violation forces the exploration of redundant edges, decreasing the possibility of reaching distant yet relevant candidates. On the other hand, graph-based ANNS usually search for relevant items greedily, which may be stuck in a narrow space after a short-length traversal among a large-scale graph. The inability to balance search depth and breadth degrades the quality of real-time recommendations. Addressing this challenge requires developing a search-compatible relevance metric and a deep-and-broad traversal algorithm for efficient graph search. *(3) Accelerating irregular neural computation in high concurrency.* At Baidu, users raise hundreds of millions of recommendation requests per second. These request executions are irregular because divergent user profiling drives distinct search paths, where the visited items should be estimated for their relevance by the neural metric. Baidu has developed a GPU-accelerated framework for web-scale recommendations to leverage the computational economy of GPUs [31]. However, their performance degrades under highly concurrent, non-uniform workloads because of frequent kernel launches, a process with fixed latency overhead that dominates runtime when computations are small and irregular [30]. Millions of sequential processes of deep relevance computations on a GPU cluster introduce bottlenecks of recommendation throughput. Therefore, the third challenge is redesigning computation scheduling to accelerate large-scale irregular neural computations for higher throughput.

To this end, we propose a **GPU-accelerated Multi-relational Parallel Graph Retrieval (GMP-GR)** framework for effective and efficient retrieval in web-scale recommendations. Specifically, we first devise a *Multi-relational User-Item Relevance Metric Learning (MUIRML)* to generate a practical DNN-based metric via multi-objective optimization. The metric extracts multi-relational knowledge in inter-dependent user behaviors to capture nuanced user preferences for accurate relevance measurement. We also infuse our metric's approximate triangle inequality property via a self-covariant loss to navigate

large graphs with shorter paths. Second, we design a *Hierarchically Parallel Approximate Nearest Neighbor Search (HiPANNS)* algorithm to achieve timely and accurate retrieval for highly concurrent queries. HiPANNS broadens the search space of the large graph item base by divergent parallel and co-optimized search processes, which can efficiently generate more relevant candidates in real-time retrieval. Based on it, HiPANNS accelerates irregular neural computations from asynchronous requests by adaptive aggregation in a GPU-based cluster for higher retrieval throughput. In Baidu recommendation applications, we further apply several system optimization strategies, including traffic dispersal, compilation optimization, model quantization, and timely stream update. Extensive experiments demonstrate the superior retrieval effectiveness and efficiency of GMP-GR in offline retrieving evaluations and online recommendation services. Our framework has been deployed in more than twenty web-scale recommender systems at Baidu, where it efficiently handles hundreds of millions of requests per second, improving user time-on-page by more than 3% for hundreds of millions of users.

The main contributions of our work are concluded as follows: (1) We propose a GPU-accelerated Multi-relational Parallel Graph Retrieval framework to achieve accurate and efficient retrieval for web-scale recommendations. (2) We devise a DNN-based metric to estimate user-item relevance via capturing inter-dependency among complex user-item relations, which can also improve the search efficiency on a large-scale graph for shorter navigation. (3) We design a GPU-accelerated parallel graph-based ANNS algorithm for highly concurrent retrieval by billion-scale breadth-depth balanced graph search and parallel irregular computations with adaptive aggregation. (4) We conduct extensive experiments to demonstrate retrieval effectiveness and efficiency improvements in offline and online recommendations at Baidu.

5.2 Preliminaries

Baidu has developed a recommendation framework as a comprehensive funnel mechanism to efficiently retrieve and rank items for users from a web-scale database for over twenty internet products (e.g., news feed, short video clips, search engine, and online advertising) [27]. The recommendation framework has served hundreds of millions of online recommendation requests per second for hundreds of millions of users. Next, we detail the retrieval framework for Baidu recommendations, including the retrieval problem and the retrieval pipeline.

5.2.1 Online Retrieval at Baidu

Given N_u users $\{u_1, u_2, \cdots, u_{N_u}\}$ and N_v items $\{v_1, v_2, \cdots, v_{N_v}\}$. Consider a query from a user u with raw feature x_u, the objective of the retrieval stage is to recommend the top-K items that are most relevant to u in order. Specifically, we define the retrieval problem as a function $\mathcal{F}(\cdot;\cdot)$ generates a set V_u based on the user features. V_u contains the K most similar items $V_u = \{v_{u,1}, v_{u,2}, \cdots, v_{u,K}\}$ as

$$V_u = \mathcal{F}(x_u; \{v_1, v_2, \cdots, v_{N_v}\}). \tag{5.1}$$

5.2.2 Retrieval Pipeline at Baidu

Baidu has constructed a graph retrieval framework in their recommendation system, including two stages: offline training and online search [29, 37].

Step 1. Offline Training

The first stage aims to learn user and item embeddings associated with a relevance metric for estimating the relevance between users and items. Embeddings and the metric are learnable modules stored by the system, which are immutable in the search stage. We can

also periodically re-train the modules once the properties change significantly. Overall, the pipeline of the training step is presented below.

(1) Users and item feature construction. We first prepare the raw features of the users and items that are denoted as $\{x_{u_1}, x_{u_2}, \cdots, x_{N_u}\}$ and $\{x_{v_1}, x_{v_2}, \cdots, x_{N_v}\}$, where N_u and N_v are number of users and items, respectively. Meanwhile, the interaction records of users u and items b are denoted as $\{D^z_{u,v}\}$. For an interaction type z, $D^z_{u_i,v_j}$ is the interactive value between a user u_i and an item v_j, which reflects how close they are in terms of the interaction z.

(2) User and item embedding projection. Then, we pre-define a user and an item projector, $\mathcal{H}_u(\cdot)$ and $\mathcal{H}_v(\cdot)$, to transform raw features into learnable embeddings as user and item representations, which are defined as $h_{u_i} = \mathcal{H}_u(x_{u_i})$ and $h_{v_j} = \mathcal{H}_v(x_{v_j})$.

(3) User-item relevance metric definition. After that, we devise a DNN-based learnable metric [35, 36] to estimate the relevance between a user and an item by a high-dimensional calculation of their embeddings. Hence, we aim to obtain a metric $\Delta(\cdot, \cdot)$ to quantify the relevance value given a user embedding and an item embedding as $\delta_{u,v} = \Delta(h_u, h_v)$.

(4) Optimization of embeddings and the metric. Next, we optimize the user embeddings, item embeddings, and the relevance metric based on the user features, item features, and user-item interactions. We describe the optimization as a function $\mathcal{F}_t(\cdot, \cdot, \cdot)$ that

$$\{h_u\}, \{h_v\}, \Delta(\cdot, \cdot) = \mathcal{F}_t(\{x_u\}, \{x_v\}, \{D^z_{u,v}\}). \tag{5.2}$$

The learned embeddings and the relevance metric are used in the online search stage to measure user-item relevance.

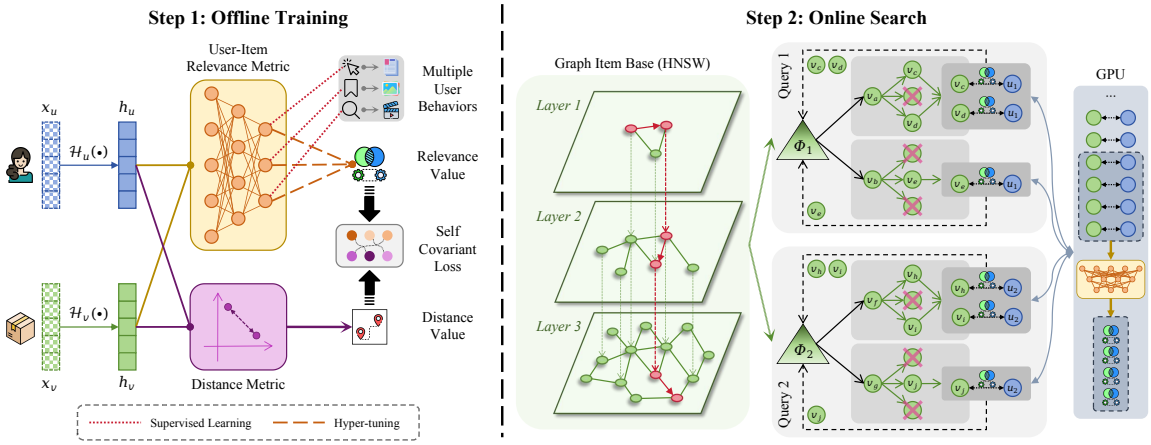


Figure 5.2: An overview of the GPU-accelerated Multi-relational Parallel Graph Retrieval framework.

Step 2. Online Search

In the second stage, we construct a graph-based item database that supports an efficient search for the target items. Traditional tree-based ANNS fails to handle web-scale data with high dimensions and uneven distributions, and requires extensive tree construction costs. Instead, we follow graph-based ANNS to construct a graph item base that can efficiently navigate possibly relevant items and retrieve satisfactory items for given users based on graph search. Specifically, we adopt HNSW [28] as the graph structure. HNSW utilizes a multi-layered graph structure, denoted as a L-layer graph G, where each upper layer contains fewer nodes to facilitate efficient search paths. Based on its hierarchy, we can effectively narrow the search space for nearest neighbors in the base layer, initiating the search process from the upper layers.

In the online search, we infer K relevant items with a querying user as follows. We perform the search iteratively across each layer of the HNSW structure, with the selected items from one layer serving as seed nodes for traversal in the subsequent layer. For each layer, we first access the corresponding user embedding. Second, we generate seed nodes, from which we start to traverse the graph. Third, we expand the candidate node set by visiting neighbors of existing nodes and continuously update the most similar ones for the

queried user. The K most similar items are ultimately extracted from the final layer. The overall search process for the relevant item set $\hat{V}_u$ for user u can be defined as

$$\hat{V}_u = \arg\max_{V_u \in V} \sum_{v \in V_u} \delta_{u,v}. \tag{5.3}$$

5.3 GPU-accelerated Multi-relational Parallel Graph Retrieval Framework

This section introduces the GPU-accelerated Multi-relational Parallel Graph Retrieval (GMP-GR) framework deployed at Baidu for web-scale recommendation. As shown in Figure 5.2, GMP-GR equips two main modules to achieve effective and efficient performance. In the training step, a multi-relational user-item relevance metric learning generates a DNN-based metric capturing inter-dependent relationships between users and items and inherits the triangle inequality property for path-efficient graph search. In the search step, a hierarchically parallel graph-based ANNS algorithm effectively delivers reliable search candidates in high concurrency, which explores a breadth-depth balanced search space and synchronizes large-scale irregular neural computations on GPUs.

5.3.1 Multi-relational User-item Relevance Metric Learning

The training step aims to optimize the user and item embeddings with the DNN-based metric to model the relevance between users and items. However, existing DNN-based metrics either fail in insufficient understanding of nuanced user preferences due to partial incorporation of multi-dimensional behaviors on items, or do not satisfy the triangle inequality property necessitated by efficient navigation of graph-based ANNS. Therefore, we propose a *Multi-relational User-Item Relevance Metric Learning (MUIRML)* algorithm to train a relevance metric that not only ensures elaborate multi-relational modeling, but also enhances navigating efficiency in the search step, ultimately improving the retrieval effectiveness.

(1) DNN-based Relevance computation. We first utilize a Multi-Layer Perceptron (MLP) to model multiple interactive behaviors. Initially, the input consists of the user embedding h_u, the item embedding h_v, and the serendipity vector $\tilde{h} \sim \mathcal{N}(0, 1)$ for diverse exploration as

$$h_{u,v}^{(0)} = \text{concat}(h_u + \tilde{h}, h_v), \tag{5.4}$$

the MLP inputs $h_{u,v}^{(0)}$ and aims to output the interaction vector $Z_{u,v}$ that stacks multiple interactions $z_{u,v}$ between user u and item v. The overall operation can be defined as

$$\hat{Z}_{u,v} = \text{MLP}(h_{u,v}^{(0)}), \tag{5.5}$$

where $\hat{Z}_{u,v}$ is the predicted interaction vector. Specifically, we define the computation of the m-th layer MLP as

$$h_{u,v}^{(m)} = \mathbf{W}^{(m)} \cdot h_{u,v}^{(m-1)} + b^{(m)}, \tag{5.6}$$

where $h_{u,v}^{(m)}$ is the m-th layer joint representation of user u and item v, and $\mathbf{W}^{(m)}$ and $b^{(m)}$ are the m-th layer learnable parameters.

Then, we leverage an attention mechanism to aggregate predicted interaction values for a joint understanding of diverse user behaviors. Specifically, the attention vector $\mathbf{A}$ is multiplied on $\hat{Z}_{u,v}$ to produce the final user-item relevance value $\delta_{u,v}$ as $\delta_{u,v} = \mathbf{A} \cdot \hat{Z}_{u,v}$ where $\mathbf{A}$ is initialized with $\frac{1}{|\mathbf{A}|}$ and can be hyper-tuned to achieve personalized behavior importance for different web recommendation scenarios.

(2) Self-supervised loss for short graph navigation. Instead of directly forcing the alignment between user-item relevance values computed by the DNN-based metric and the Euclidean distance, we aim to retain the triangle inequality property by transplanting the ranking regularity of Euclidean distance into the DNN-based metric. Specifically, we devise a Self Covariant Loss (SCL) to quantify the gap of the ranking regularity via the divergence of the pairwise values covariance for two metrics. By minimizing the loss, the order of

items ranked according to our metric will be similar to the Euclidean distance, while the relevance values will be maximally retained for accurate estimation by multi-dimensional understanding. In this way, we can eliminate unnecessary traversal paths and increase the possibility of approaching the optimal candidates.

Specifically, we randomly sample N_l pairs of users and items $\{(u_{i_1}, v_{j_1}), \cdots, (u_{i_{N_l}}, v_{j_{N_l}})\}$ and compute their DNN-based relevance values $\{\delta_{u,v}\}$ and Euclidean distance $\{\delta'_{u,v}\}$, respectively. Then, we calculate the correlation of two groups of values based on their covariance as

$$\rho_{u,v} = \frac{\sum_{k=1}^{N_l} (\delta_{u_k,v_k} - \overline{\delta_{u,v}})(\delta'_{u_k,v_k} - \overline{\delta'_{u,v}})}{\text{VAR}(\delta_{u,v}) \cdot \text{VAR}(\delta'_{u,v})}, \tag{5.7}$$

where $\text{VAR}(x) = \sqrt{\sum_{k=1}^{N_l}(x - \overline{x})^2}$.

(3) Multi-objective optimization for embeddings and the relevance metric. Lastly, we design the multi-objective optimization to train the user and item embeddings, as well as the DNN-based relevance metric. The loss is defined as

$$\mathcal{L} = \sum_{i=1}^{N_u} \sum_{j=1}^{N_v} (\| Z_{u_i,v_j} - \hat{Z}_{u_i,v_j} \|_2 + \sum_{k=1}^{N_l} \frac{1}{\rho_{u_{i_k},v_{j_k}} + 1}), \tag{5.8}$$

which will be optimized by gradient-based algorithms. In this way, we can simultaneously minimize the prediction error of multi-relational interaction behaviors and maximize the similarity of ranking regularity between our DNN-based metric and Euclidean distance for triangle inequality property, which can generate shorter paths in graph search for more efficient retrieval.

5.3.2 GPU-accelerated Hierarchically Parallel Approximate Nearest Neighbor Search

In this work, the search step aims to retrieve the most similar items using graph-based ANNS. Existing graph retrieval frameworks struggle to achieve low-latency retrieval in high-concurrency scenarios for two reasons. First, their traversal strategies underperform at scale, as they inadequately balance search depth, *i.e.* refinement of local neighborhoods, and breadth, *i.e.* coverage of global graph regions, within constrained computational budgets [61]. Second, the underlying GPU-based infrastructure faces inefficiencies when handling asynchronous retrieval requests, which includes tremendous irregular neural computations according to divergent search routes. The frequent kernel launch of GPU computation from different queries induces significant overhead and contention under extreme concurrency [82]. These limitations collectively degrade the retrieval throughput for real-time recommendations. To this end, we propose a *Hierarchically Parallel Approximate Nearest Neighbor Search (HiPANNS)* algorithm to efficiently answer highly concurrent retrieval queries. Specifically, HiPANNS consists of two sub-algorithms, *i.e.* inter-candidate HiPANNS and inter-query HiPANNS.

Inter-candidate HiPANNS (C-HiPANNS) allows parallel search within the large-scale graph item base to broaden the global search space for more comprehensive retrieval. For each query u, we aim to execute k parallel search processes with different initial seeds S. Each search process traverses the graph item base by visiting the most relevant neighbors. We jointly maintain candidates by a shared max heap Φ_u that ranks k current most relevant items among processes. In this work, we carry out the multi-processing at each layer of the HNSW graph, seeded with top-k candidates ranked by Φ_u of the last layer. The parallelized traversal achieves a k-fold expansion of the search space, which yields more effective retrieval results.

Inter-query HiPANNS (Q-HiPANNS) adaptively aggregates irregular neural computations for more efficient GPU-accelerated retrieval against launch bound from extreme concurrency. We adaptively synchronize the irregular user-item relevance evaluation by a queuing aggregation scheme and parallelize batched neural computation on GPU for acceleration. First, we initialize a queue Ψ to persistently listen to neural computations. Ψ will record the computations orderly for the following sequential batching, *i.e.* the element received first will be batched first. Each element in Ψ is produced from C-HiPANNS as a group of (u, v) pairs for measuring neighbor embedding similarity. We dynamically generate batches for parallel computing the GPU cluster U. However, the elements of Ψ vary in amounts due to divergent connections in the graph item base, causing difficulties in aggregating fixed batches that are beneficial to fully utilize the computational ability of GPUs. To address that, we synthesize batches by adaptively slicing (u, v) groups to fit certain configurations. Specifically, we set the maximum size of the computing batch as B_U. If the size of current batch B_i, N_{B_i}, is larger than the remaining space ϵ, we split it into two elements, including $B_i^{[1,\epsilon]} = \{(u_{i,1}, v_{i,1}), \cdots, (u_{i,\epsilon}, v_{i,\epsilon})\}$ and $B_i^{[\epsilon+1,N_{B_i}]} = \{(u_{i,\epsilon+1}, v_{i,\epsilon+1}), \cdots, (u_{i,N_{B_i}}, v_{i,N_{B_i}})\}$. $B_i^{[1,\epsilon]}$ will be computed in the current executive round of U while $B_i^{[\epsilon+1,N_{B_i}]}$ will be computed in the next. The output values will be returned as estimations of the relevance between users and items during the processes of C-HiPANNS.

More details are presented in Appendix C.1.

5.4 System Optimization

We devise various system optimization strategies to accelerate our retrieval framework deployed at Baidu.

Traffic Dispersal We construct a communicative computing cluster composed of multiple interconnected servers designed to facilitate traffic dispersal and mitigate the effects of

Table 5.1: Offline retrieval performance of different methods on four datasets for three metrics.

	Video			MVideo		
	Cov	Rec@10	Rec@100	Cov	Rec@10	Rec@100
C-NSW	0.3386	0.3829	0.4028	0.3185	0.3775	0.3986
D-NSW	0.4084	0.5527	0.6094	0.4283	0.5428	0.5908
SL2G	0.4413	0.5784	0.6121	0.4428	0.5684	0.6049
NANN	0.5368	0.6717	0.6635	0.5021	0.6491	0.6498
GMP-GR	**0.6492**	**0.8239**	**0.8052**	**0.6044**	**0.7175**	**0.7325**
	Explore			Fresh		
	Cov	Rec@10	Rec@100	Cov	Rec@10	Rec@100
C-NSW	0.3288	0.3486	0.3674	0.3493	0.3823	0.4084
D-NSW	0.4081	0.5265	0.5376	0.5083	0.6075	0.6344
SL2G	0.4076	0.5249	0.5485	0.5123	0.6285	0.6298
NANN	0.5029	0.6038	0.6345	0.6130	0.7318	0.7555
GMP-GR	**0.6279**	**0.7999**	**0.7810**	**0.6906**	**0.8420**	**0.8428**

extreme query loads. Each server is equipped with a monitoring module that continuously assesses its computing load by analyzing available memory and current query traffic levels. When a server reaches its saturation point, it proactively requests assistance from its peer servers to redistribute the excess queries. The receiving servers will accept these queries if they possess adequate computational resources at that time. This approach significantly enhances overall system efficiency by optimizing the utilization of distributed computational resources.

Compilation Optimization Metric learning and graph search algorithms employ fundamental operators. To accelerate retrieval processes, we integrate these operators to minimize redundant computations based on existing techniques of computational graph optimization and pattern recognition [83]. Our system includes an automatic analysis module for operator kernel compilation, which aids in the optimization of neural computations for specific hardware configurations. Moreover, to address the launch bound issue that may affect computational performance on GPUs [84], we abstract the execution process of DNN-based

operations as a graph. This approach utilizes an adaptive bucketing mechanism to aggregate independent submissions of individual operations into a single, cohesive submission, thereby enhancing computational efficiency.

Model Quantization We utilize a post-training quantization technique on the DNN-based relevance metric to decrease the computing costs with little accuracy loss. Specifically, we reduce the size of the entire MLP connection layers by quantizing the weights from 32-bit floating point numbers to 16-bit. After reductions, the metric only requires half of the size for computing and memorization and supports faster operators tailored for lower precise calculation. In large-scale recommendations, this method can reduce retrieval latency and significantly conserve computational resources on participating processing units for higher throughput.

Timely Stream Update The user and item information is time-varying. To deliver accurate recommendations over time, it is crucial for the system to maintain up-to-date user and item representations and model parameters to reflect the evolving knowledge. However, increasing the freshness of this information necessitates more frequent updates, which can lead to higher training costs. To balance freshness and training costs, we first design a stepwise entry scheme to select active users and items via a dynamic threshold for timely updates. Besides, we use a click trigger that activates updates of the item base to incorporate items that occurred initially. Fresh data is then cumulatively integrated with the original item base in a batched manner to ensure both responsiveness and efficiency in handling data updates.

5.5 Experiments

5.5.1 Experimental Setup

We present datasets, baselines, and metrics in our offline and online experiments. Besides, implementation details are provided in Appendix C.1.1.

Datasets. We collect real-world video resources from Baidu applications and construct four datasets according to their potential of user satisfaction, including (1) Video: popular landscape videos, (2) MVideo: popular portrait videos, (3) Explore: videos that are potentially favorable but not popular, and (4) Fresh: videos that are uncommon and infrequently discovered.

Baselines. We adopt four retrieval baselines in offline experiments, including (1) C-NSW: a basic retrieval framework with cosine function as the relevance metric based on NSW [85]; (2) D-NSW: a DNN-based retrieval framework based on NSW, supervised by single user-item interaction; (3) SL2G [37]: a DNN-based retrieval framework based on HNSW [28]; and (4) NANN [29]: an industrial retrieval framework improved on SL2G with adversarial training and more efficient ANNS.

Metrics. For offline experiments, we involve three metrics in our evaluations on retrieval performance, including Coverage (Cov), Recall@10 (Rec@10), and Recall@100 (Rec@100). We take the brute-force-sorted items as the ground truths of retrieval. Cov describes the percentage of retrieved items that rank in the top 100 items of the ground truths. Rec@10 and Rec@100 measure the percentage of the top 10 and 100 items of the ground truths in the top 10 and 100 retrieved items, respectively. Moreover, we evaluate the speed by the number of items retrieved per second. For online testing,

Table 5.2: Retrieval speed (items per second).

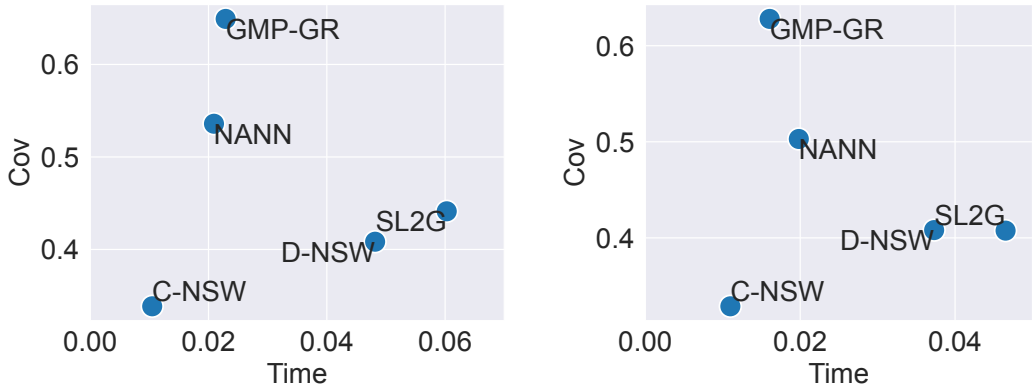
	Video	MVideo	Explore	Fresh
C-NSW	520,390	405,096	521,639	345,577
D-NSW	163,093	163,821	199,163	170,889
NANN	281,758	226,280	248,118	150,022
SL2G	134,947	145,312	162,033	176,316
GMP-GR	321,352	299,242	384,458	229,902

we collect query traffic, latency, and throughput of servers for different services across timestamps.

5.5.2 Offline Evaluation

We evaluate the retrieval performance and efficiency of GMP-GR and baseline frameworks on four datasets in an offline environment.

Retrieval Performance. Overall, GMP-GR consistently outperformed the baseline methods across all settings. For instance, GMP-GR demonstrated improvements of at least 20.94%, 22.66%, and 21.36% in terms of Cov, Rec@10, and Rec@100, respectively, on the Video dataset. Notably, significant improvements are also evident on other datasets. Specifically, compared to SL2G and NANN, two typical industrial retrieval frameworks consistently based on HNSW, GMP-GR produces more accurate retrieved results. GMP-GR achieved performance improvements of 20.37%, 24.86%, and 12.66% over SL2G and NANN on Cov on MVideo, Explore, and Fresh, respectively. The potential reasons are that (1) GMP-GR overcomes the shortage of SL2G that the neural metric may harm graph search efficiency due to the unsatisfaction of triangle inequality, and (2) GMP-GR incorporates valuable insights from dimensional interactive information in metric learning and leverages a broader search space for higher possibility of effective retrieval.

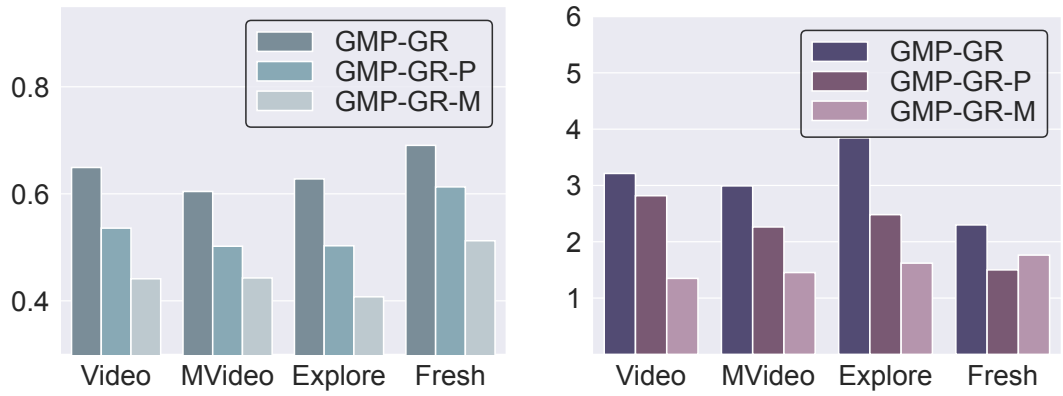


(a) Video

(b) Explore

Figure 5.3: Analysis of correlations between retrieval performance and efficiency on Video and Explore datasets. Methods close to the left top (↖) have better retrieval performance, *i.e.* Cov (↑), with less time costs (←).

Efficiency Analysis. This part analyzes the efficiency by testing their inference time and search speed. We first analyze the correlation between retrieval performance and time by scattering five methods in Figure 5.3. More results on MVideo and Fresh datasets are in Appendix C.1.2. Intuitively, a method is considered more efficient if it achieves higher performance (here we use Cov to evaluate) with lower time consumption. Results reveal that GMP-GR outperforms the four baseline methods in terms of efficiency, as evidenced by its proximity to the upper left quadrant in the scatter plots. Notably, C-NSW stands out for its minimal retrieval time, attributed to its reliance on the cosine similarity metric which is computationally lightweight than DNNs. However, C-NSW underperforms in retrieval accuracy due to the limitation that the simple metric is unable to capture complex user-item relationships. Then, we compare the retrieval speed in Table 5.2, which reflects the number of nodes that methods traverse on HNSW per second. GMP-GR also exhibits the capability to search through more nodes (*i.e.* larger search space) than any DNN-based approach. In summary, GMP-GR consistently outperforms all baseline methods, striking an optimal balance between retrieval efficiency and accuracy.



(a) Cov

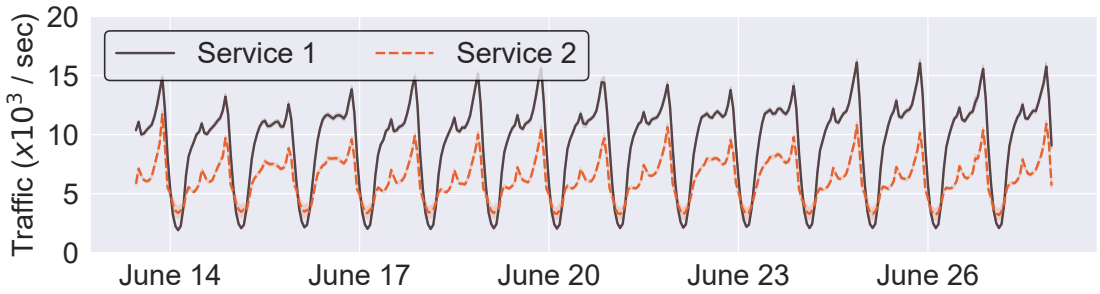
(b) Speed (10^6 items / sec)

Figure 5.4: Ablation study of GMP-GR on retrieval performance (Cov) and speed.

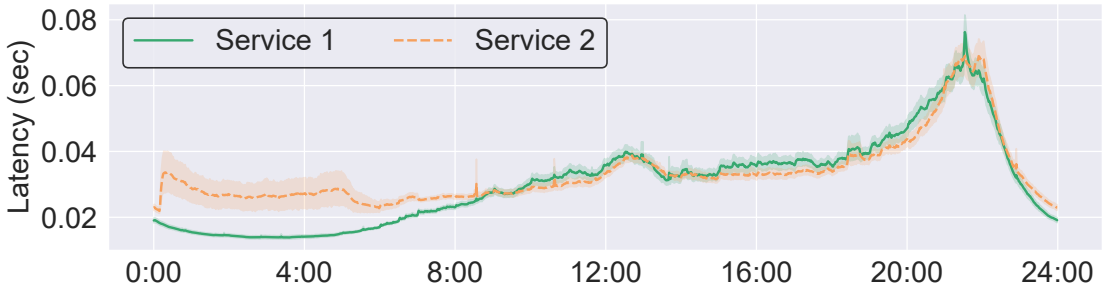
Ablation Study. We conduct an ablation study on GMP-GR in Figure 5.4, by comparing it with two variants, *i.e.* GMP-GR-P and GMP-GR-M. Evaluations on Rec@10 and Rec@100 are provided in Appendix C.1.3. GMP-GR-P replaces MUIRML with a single supervision of the behavior in which a user visits an item, and GMP-GR-M substitutes HiPANNS with a standard HNSW search algorithm. Specifically, we find that GMP-GR consistently outperforms both GMP-GR-P and GMP-GR-M across different datasets. This superiority can be attributed to two key factors. First, MUIRML significantly enhances the relevance estimation accuracy and pathfinding efficiency. Second, HiPANNS prevents the graph-based ANNS process from succumbing to local optima by distributed and co-optimized search. In addition, GMP-GR also shows a notable advantage in retrieval speed that replacing HiPANNS with a standard HNSW search algorithm can result in a great degradation of retrieval speed. This observation underscores the pivotal role of HiPANNS in acceleration. Furthermore, a more effective user-item relevance metric not only enhances the accuracy but also reduces traversal steps to converge to optimal candidates.

Table 5.3: Online user statistics for two services.

	Time on Page		Daily Active Users	
	Service 1	*Service 2*	*Service 1*	*Service 2*
GMP-GR	+1.13%	+1.19%	+0.23%	+0.50%



(a) Traffic of two weeks in June, 2024.



(b) Latency in one day.

Figure 5.5: Temporal statistics of traffic and latency.

5.5.3 Online Testing

We deploy GMP-GR in the Baidu Mobile APP for two services supported by five servers for online testing. Service 1 is for the main recommendation list on the initial page. Service 2 is for the next recommendation sequence that can be accessed after clicking on a video in immersive mode.

Time on Page and Daily Active Users

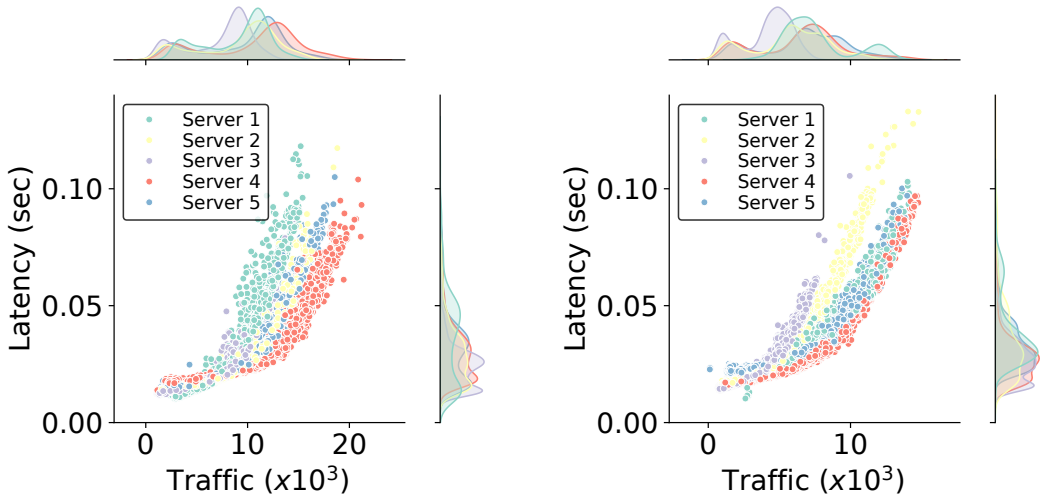
Table 5.3 depicts the impact of deploying GMP-GR on time on page and daily active users. Our results indicate a significant improvement in time on page and daily active users across the two services. The integral components of GMP-GR, MUIRML, and HiPANNS, contribute to a 0.57% and 0.23% improvement in time on page, respectively. Our findings suggest that the deployment of GMP-GR with improved modules has positively impacted user engagement and satisfaction.

Query Throughput

We also examine the query throughput influenced by the system acceleration strategies. The results demonstrate a strong improvement in query throughput, with an approximate doubling of throughput in the two services. Specifically, our analysis shows that the compilation optimization has led to a 74% and 75% increase in throughput by simplifying the computational graphs. Similarly, the model quantization has resulted in a 26% and 25% increase in throughput by decreasing DNN-based calculations. These findings underscore the effectiveness of GMP-GR in optimizing query processing and system efficiency.

Statistical Analysis of Traffic and Latency

We collect traffic and latency statistics per second, which are distributedly recorded at five servers in two services. First, we examine the temporal patterns as shown in Figure 5.5. Specifically, Figure 5.5a illustrates the traffic trends over a two-week period, highlighting periodic fluctuations in query volumes. Meanwhile, Figure 5.5b provides a more granular view of latency statistics over a 24-hour cycle, revealing variations in online response times in one day. Moreover, we investigate the distributions of traffic and latency and explore their



(a) Service 1 (b) Service 2

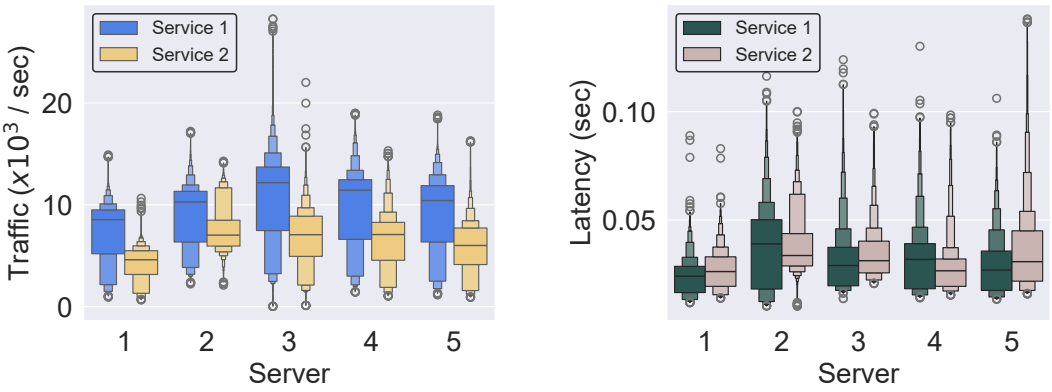
Figure 5.6: Analysis of distributions of traffic and latency.

correlations in Figure 5.6. The result indicates a direct relationship between query traffic and latency, with higher traffic volumes leading to longer response times. Furthermore, we observe distinct patterns in the latency-traffic relationships across five servers, reflecting divergence in computational resources and workload distribution. Finally, we summarize the traffic and latency statistics across five servers in Figure 5.7, providing insights into the performance characteristics in terms of query processing and response times.

5.6 Related Works

This section reviews related works, including DNN-based recommender systems and approximate nearest neighbor search.

Deep Neural Network-based Recommender Systems Deep Neural Network (DNN) has become a popular tool for recommendation systems due to its ability to capture intricate user-item interaction patterns through nonlinear transformations [80]. For example, [86] leverages convolutional and recurrent neural networks for recommendations to learn text



(a) Traffic (b) Latency

Figure 5.7: Traffic and latency distributions.

features from tweets. Recently, neural collaborative filtering has been increasingly adopted, where a Multi-layer Perceptron (MLP) is deployed to approximate the interaction function and has significant performance gains over traditional methods [87]. Along with the idea, two-tower architectures have become popular in DNN-based recommender systems [34]. They train dual encoders for users and items, respectively, which generate user and item representations for downstream operations such as relevance evaluation. With two sets of high-dimensional embeddings, the DNN-based system can provide recommendations in user-based or item-based paradigms [88]. User-based collaborative filtering compares similarities between users based on their past ratings and recommends items liked by similar users [89]. In contrast, item-based collaborative filtering computes similarities between items rather than between users [90]. We design a DNN-based metric optimized in a multi-objective way to incorporate comprehensive behaviors and triangle inequality for better recommendation accuracy and efficiency.

Approximate Nearest Neighbor Search Approximate Nearest Neighbor Search (ANNS) is a fundamental problem in computer science, particularly within domains that handle high-dimensional data such as computer vision [91] and multimedia [92]. ANNS aims to

find points in a dataset that are close to a query point, where searching the exact nearest neighbor is computationally prohibitive [93]. One of the early approaches, KD-tree [81], performs well in low-dimensional spaces. But their efficiency decreases significantly as dimensionality increases, called ”curse of dimensionality”. To address the limitation, more sophisticated tree-based structures like Randomized KD-trees [94] attempt to partition the data space more effectively and offer improved performance in higher-dimensional spaces. As research progressed, hashing-based methods emerged as a powerful alternative for ANNS, with Locality-Sensitive Hashing (LSH) being a prominent technique to hash input items by mapping similar items to the same ”buckets” with high probability [32]. More recently, graph-based methods have gained popularity for their effectiveness in handling high-dimensional ANNS [33]. The idea is to construct a graph where nodes represent the dataset points, and edges connect points close to each other. Navigating this graph allows for efficient ANN searches. Notably, the Navigable Small World (NSW) [85] and its improved variant, Hierarchical Navigable Small World (HNSW) [28], have shown remarkable performance, significantly outperforming previous methods on large-scale datasets. Besides, instead of Delaunay graphs, other graph-based approaches choose K-nearest neighbor graph [59] and relative neighborhood graph [60] for index construction. In industrial applications, SL2G [37] first leverages DNN as the relevance metric for graph-based ANNS in real-world recommendations. NANN [29] extends SL2G by improving the DNN-based metric for better accuracy and adaptivity and working ANNS with controllable costs for industrial requirements. In this work, we propose a parallel retrieval algorithm for web-scale recommendations with wider search space and better concurrent computational efficiency.

5.7 Conclusion

This work investigates the retrieval stage optimization for the web-scale recommendation. We propose a GMP-GR framework to enhance the retrieval accuracy and efficiency. First, we devise multi-relational user-item relevance metric learning to improve user-item multi-relational understanding and graph search efficiency in a multi-objective way. Second, we design a tailored parallel graph-based ANNS algorithm to distributedly broaden search space and parallelize irregular neural computations on GPUs. Besides, we employ system optimization strategies to improve online recommendation performance. We conduct extensive experiments on offline datasets and online services to demonstrate the outperformance of GMP-GR. We have implemented GMP-GR in the recommender systems at Baidu, where it serves hundreds of millions of users and processes hundreds of millions of online requests per second.

CHAPTER 6

CONCLUSION AND FUTURE DIRECTIONS

6.1 Conclusions

Modern video retrieval must operate across heterogeneous query forms, long and informationally dense videos, and candidate collections whose scale precludes exhaustive scoring. This thesis has argued that these demands are best understood as three sequential scaling challenges along the retrieval pipeline: semantic scaling, observational scaling, and search scaling. Each presents a distinct bottleneck that is not eliminated by optimizing an upstream stage alone.

The first study (Chapter 3) diagnosed capability gaps in fine-grained, temporal, and partially relevant retrieval across sixteen public datasets, then addressed them through a synthesis pipeline producing 1.55 million multi-task video-text pairs and a curriculum learning strategy using inherent task hierarchies. The resulting model achieved state-of-the-art zero-shot performance across diverse retrieval scenarios, and the evaluation reveals that partially relevant retrieval is a pivotal yet overlooked real-world challenge. The second study (Chapter 4) introduced the first multi-granular long-video retrieval benchmark and found, across twenty-three multimodal embedding models, that fine-grained retrieval lags coarse-grained performance by over 75% under sparse sampling. This finding shows that observational scaling is a structural bottleneck rather than a secondary preprocessing choice. A dense-to-sparse inference framework built on this analysis achieved up to 20% fine-grained improvement while operating 4$\times$ to 16$\times$ more efficiently. The third study (Chapter 5) proposed a GPU-accelerated graph retrieval framework with co-designed multi-relational

metric learning, parallel graph search with adaptive GPU aggregation, and system-level optimizations including traffic dispersal, compilation optimization, and model quantization. Deployed across more than twenty applications at Baidu, the system serves hundreds of millions of users at hundreds of millions of requests per second, improving user time-on-page by more than 3%. This study shows that learned retrieval signals must be co-designed with index structures and hardware execution, since retrieval accuracy alone is insufficient for deployment-scale search.

Across all three studies, a central finding is that scaling capacity alone, whether model size, observation density, or metric accuracy, does not automatically yield broader retrieval generalization unless aligned with the specific bottleneck at each pipeline stage. Generalizability serves as the empirical criterion for assessing whether learned capability transfers beyond the conditions under which it was optimized. In semantic scaling, transfer is assessed across query modalities, retrieval domains, and relevance granularities. In observational scaling, it is assessed across sampling densities, video durations, and query granularities. In search scaling, it is assessed across deployment contexts, traffic patterns, and serving constraints. These forms differ in their specifics but converge on the same principle: scaling is meaningful only when the resulting capability extends beyond the narrow condition under which it is optimized.

6.2 Future Directions

Each study in this thesis also points toward capabilities that extend beyond the present work. This section outlines four directions for next-generation video retrieval systems, suggesting that video retrieval may evolve from a standalone embedding task into a foundational capability for multimodal intelligence.

6.2.1 Retrieval-Native Multimodal Foundation Models

Current multimodal large language models are typically trained for generation or instruction following, and retrieval is added afterward through contrastive fine-tuning or a separate retriever. This creates a tension: hidden states optimized for next-token prediction differ fundamentally from the compact, discriminative representations that retrieval requires. A promising direction is to design foundation models that are retrieval-native rather than retrieval-adapted, exposing retrieval-compatible hidden states through mechanisms such as retrieval-specific tokens, task-decoupled routing, or lightweight adapters, while preserving generative and instruction-following abilities. The deeper goal is a retrieval-compatible interface inside the foundation model, so that dense embeddings inherit the model's world knowledge and multimodal reasoning. An open problem is how to achieve this without degrading generation quality.

6.2.2 Video Retrieval as Long-Term Memory for Interactive and Embodied Agents

Current benchmarks treat videos as fixed inputs, but deployed systems increasingly process continuous visual streams. Future retrieval systems may need to operate as streaming visual memory, continuously encoding incoming video and supporting real-time queries against a growing memory store. An agent in a dynamic environment may need to retrieve the last observed location of an object, recall relevant visual context from the past few hours, or detect anomalous events within a recent time window. These tasks require retrieval over temporally distributed evidence under strict latency constraints, connecting observational scaling to search scaling. Evaluation would shift beyond Recall@K toward downstream task success, decision latency, memory usefulness, and the ability to retrieve sparse events from continuous streams. In such systems, video retrieval becomes a memory access mechanism embedded in the perception-action loop of multimodal agents.

6.2.3 Self-Evolving Data, Evaluation, and Curriculum Optimization

The evaluation-guided synthesis strategy in Chapter 3 shows that targeted data generation can close specific capability gaps more effectively than indiscriminate scaling. This principle can be extended into a self-evolving loop: an evaluator diagnoses weaknesses across fine-grained retrieval capabilities, a synthesizer generates targeted multimodal training examples, and a curriculum scheduler dynamically adjusts training distributions based on evaluation feedback. The updated model is then re-evaluated, closing the loop. Such a framework could reduce reliance on manually designed datasets and curricula, enabling retrieval models to continuously adapt to new query forms, video domains, and relevance patterns that static benchmarks cannot fully anticipate.

6.2.4 Continuous Generative Representations for Multi-Granular Video Retrieval

Video semantics are inherently multi-granular, and a single pooled embedding may be insufficient for queries spanning different semantic resolutions. Diffusion and flow-based generative models transform representations through continuous trajectories, where different stages may expose representations at different levels of abstraction. Early denoising steps may capture global scene structure, while later steps may encode fine-grained action and temporal details. Future work could explore extracting retrieval embeddings from intermediate generative states, enabling a single model to produce multi-granular representations. This direction remains speculative: significant challenges remain in making generative trajectories discriminative for retrieval, and it is an open question whether such models can match purpose-built encoders at comparable cost.

Together, these directions suggest that video retrieval may evolve from a standalone embedding task into a foundational interface for multimodal intelligence, extending the thesis

beyond scalable retrieval systems toward retrieval-centered multimodal systems that can remember, adapt, and reason over visual experience.

CHAPTER 7

LIST OF INVOLVED PUBLICATIONS

This thesis is based on the following three first-authored publications. Each publication corresponds to one of the three main research chapters of this dissertation, addressing the semantic, observational, and search scaling challenges of video retrieval, respectively.

1. **Chapter 3** — Zhuoning Guo, Mingxin Li, Yanzhao Zhang, Dingkun Long, Pengjun Xie, and Xiaowen Chu. "Towards Universal Video Retrieval: Generalizing Video Embedding via Synthesized Multimodal Pyramid Curriculum." Submitted to *Proceedings of the 34th ACM International Conference on Multimedia (MM '26)*. ACM, 2026.

2. **Chapter 4** — Zhuoning Guo, Tingyu Song, Mingxin Li, Yanzhao Zhang, Dingkun Long, Pengjun Xie, Xiaowen Chu. "Demystifying Spatiotemporal Scaling Laws of Multimodal Video Embeddings." Submitted to *Proceedings of the European Conference on Computer Vision (ECCV 2026)*. Springer, 2026.

3. **Chapter 5** — Zhuoning Guo, Guangxing Chen, Qian Gao, Xiaochao Liao, Jianjia Zheng, Lu Shen, and Hao Liu. "GPU-accelerated Multi-relational Parallel Graph Retrieval for Web-scale Recommendations." In *Proceedings of the ACM Web Conference 2026 (WWW '26)*. ACM, 2026.

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